

Supplementary Material — Software-as-a-Graph: Parameter Sensitivity of the Explanation Layer

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S1. Notation and Architectural Concepts

Table S1 summarizes the mathematical symbols, graph representations, and evaluation metrics used across the main manuscript and this supplementary document.

Table S1: Notation used throughout the paper.

Symbol	Description
$G_{\text{structural}}$	Raw multigraph (used exclusively by simulation oracles)
G_{analysis}	Logical DEPENDS_ON projection (input to predictors)
V_{app}	Application nodes (the primary scored population)
$w(t), w(e)$	QoS topic weight; derived dependency edge weight ($w(e) = w_E(e)$ on DEPENDS_ON)
$I^*(v)$	Primary cascade-reachability simulation oracle
$\hat{I}_1^*(v)$	Analytic first-order closed-form expansion of I^* (Analytic I^*)
$I_R^*(v)$	Reliability-scaled feed loss (auxiliary prediction target)
$I_{\text{dyn}}(v)$	Dynamic queue-flow discrete-event simulation oracle ($N = 1,321$ full population)
$I_{\text{dyn}}^{n=30}$	Exploratory developmental sample of queue-flow simulation ($n = 30$ per fold)
$I_{\text{comp}}(v)$	Multi-criteria composite simulation oracle
RL, FR	Reachability Loss and Fragmentation components of I_{comp}
TL, FD	Throughput Loss and Flow Disruption components of I_{comp}
ρ	Spearman rank correlation coefficient (full population)
$\rho_{>0}$	Spearman rank correlation restricted to active stratum ($I > 0$)
Overlap@ K	Top- K identification set overlap ($K = 0.20 V_{\text{app}} $)

S2. Entity Weights and the Infrastructure Dependency Rules

Entity weights. An Application's w_V is the power mean ($p = 3$) of the QoS weights $w(t)$ of its topics. A Library ℓ takes the largest weight among its topics $T(\ell)$ and consuming Applications $C(\ell)$, amplified by its fan-out:

$$w_V(\ell) = \min\left(1, \max\left(\{w(t)\}_{t \in T(\ell)} \cup \{w_V(a)\}_{a \in C(\ell)}\right) \cdot \left(1 + 0.15 \log_2(1 + |C(\ell)|)\right)\right).$$

These weights enter the QoS-weighted node features and the Rule 5 edge weight. The dependency counts reported in the main text are unweighted, and declared QoS policies carried no measurable signal on either the reachability or the queue-flow simulator (main text, Section 3.2).

Infrastructure rules. Table S2 lists the four DEPENDS_ON rules for brokers and hosts that complete the derivation for system-level analysis. None is exercised by the Application-level evaluation of the main text, which ranks Applications on the Rules 1 and 5 projection; evaluating them by ranking Brokers and Hosts is future work (main text, Section 7.5). Rule 2 combines the topics a component shares with a broker by probabilistic union, as Rule 1 does.

Table S2: Infrastructure DEPENDS_ON rules (not evaluated).

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
2	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3	host_to_host	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	$\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_E(u \rightarrow v)$
4	host_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	$\max_{u \in \text{hosted}(h)} w_E(u \rightarrow b)$
6	broker_to_broker	Broker $\leftrightarrow$ Broker (shared fault-domain colocation, symmetric)	$w_V(\text{host})$

S3. Heterogeneous Graph Transformer Message-Passing Formulation

For completeness, we record the layer-wise equations of the Heterogeneous Graph Transformer (HGT) [1] implemented via PyTorch Geometric’s HGTConv [2] and summarized in Section 4.1 of the main manuscript. For source node u and target node v connected by meta-relation $\tau(e) = (\tau(u), \phi(e), \tau(v))$:

1. **Type-Specific Projection:** Raw node features x_v (dimension 19–25) are mapped into hidden dimension D :

$$h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v)) \quad (1)$$

2. **Relational Mutual Attention:** For attention head $i \in \{1, \dots, H\}$, with incoming neighborhood $\mathcal{N}(v)$ and incorporated edge representation $\tilde{h}_v = h_v + e'_{uv}$:

$$\text{Attn}^i(u, e, v) = \text{Softmax}_{u \in \mathcal{N}(v)} \left(K^i(u) W_{\text{att}, \phi(e)}^i Q^i(\tilde{h}_v)^\top \cdot \frac{\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}}{\sqrt{D/H}} \right) \quad (2)$$

where $\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}$ is the learned per-meta-relation scaling prior (`p_rel` in PyG), weighting entire relation triples independently of individual node embeddings.

$$\text{Msg}(u, e, v) = V(u)W_{\text{msg}, \phi(e)} \quad (3)$$

3. **Residual Aggregation and Layer Normalization:** For layers $l \in \{1, \dots, L\}$:

$$h_v^{(l)} = \text{LayerNorm} \left(h_v^{(l-1)} + \text{Dropout} \left(\sum_{u \in \mathcal{N}(v)} \text{Attn}(u, e, v) \cdot \text{Msg}(u, e, v) \right) \right) \quad (4)$$

S4. Parameter Sensitivity of the Explanation Layer

This section concerns the proposed explanation layer (Section S27), not any ranker evaluated in the main text. Its RM composite scores each component’s Reliability and Maintainability risk from structural and code metrics; Section S27 gives its formulas.

Three things follow. First, the parameter risk of the explanation layer is concentrated in two constants, both internal to the RM composite; the QoS-derived weights on which the framework’s architectural story rests are not load-bearing for its output. Second, the topic-weight split $(\beta, \alpha, \psi) = (0.75, 0.15, 0.10)$ is a documented convention rather than a tuned parameter — no choice within its simplex, including uniform weighting, would meaningfully change any result in this paper. Third, Dirichlet sampling over 100 draws of the whole weight vector confirms that the parameterization is stable as a whole: mean $\rho = 0.200$ (sd 0.010, 90% interval $[0.185, 0.218]$), with mean Kendall $\tau = 0.826$ and mean top-20% Jaccard 0.739 against the shipped ranking.

Direction of the shrinkage effect. The λ row deserves separate comment because its direction is unfavorable. Rank correlation falls monotonically as the intra-dimension weights move from a uniform prior toward the raw AHP judgment ($0.319 \rightarrow 0.200$, Figure S1): expert elicitation makes the ranking worse, by 0.119. We retain $\lambda = 0.70$ regardless, and the reasoning should be explicit rather than assumed. RM is an attribution instrument, not a ranking model (Section S27); its purpose is to say *why* a component is fragile in auditable, standards-referenced terms, and rank correlation against a cascade oracle is not the quantity it is built to

Table S3: Sensitivity of the RM composite to its ten declared weight constants. **OFAT** reports one-factor-at-a-time sweeps against $I^*(v)$ over the seven core synthetic domains; **Morris** reports elementary-effects screening over six scenarios (10 trajectories, 110 evaluations), where μ^* is influence on mean ρ and σ its interaction spread. Rows are ordered by μ^* . Only r_α and λ are load-bearing.

Constant	One-factor sweep (OFAT)	Morris μ^*	Morris σ
r_α (Fault Tolerance / Availability blend)	not swept individually	0.132	0.040
λ (AHP shrinkage, uniform $\rightarrow$ raw)	$\rho = 0.319 \rightarrow 0.200$, monotone (spread 0.119)	0.134	0.090
w_{dur} (QoS durability)	part of the AHP-derived QoS vector (0.24, 0.62, 0.14); not swept separately	0.013	0.016
w_{prio} (QoS priority)		0.022	0.024
w_{rel} (QoS reliability)		0.012	0.018
α (topic payload size)	swept jointly over the full (β, α, ψ)	0.019	0.026
β (topic QoS term)	simplex, 7 points: $w(t)$ ordering never	0.025	0.028
ψ (topic frequency)	falls below $\rho = 0.919$ against the	0.011	0.011
p (power-mean exponent)	shipped split: downstream spread	0.005	0.007
γ (library fan-out)	not swept individually 0.031 (Topo-QoS), 0.007 (RM) not swept individually	0.001	0.001

maximize. Tuning λ toward zero would improve a number we do not claim at the cost of the traceability we do. What this sweep establishes is nonetheless a genuine limitation: we have evidence that the elicited weights produce worse rankings and none that they produce better attributions. Validating the attribution itself — against practitioner judgment, or against the outcome of applying the repairs it recommends — is outside this study and is the explanation layer’s principal open question (Section 7.5 of the main manuscript).

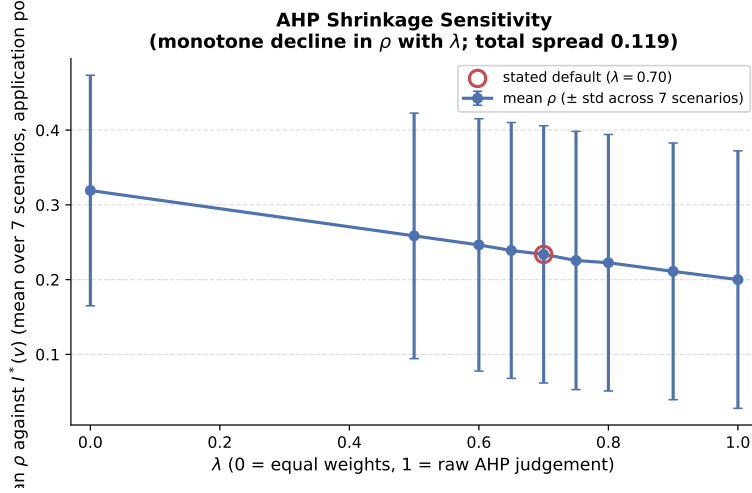


Figure S1: RM composite rank correlation against $I^*(v)$ across the AHP shrinkage parameter λ (Application population, mean over the seven core synthetic domains). The descent is monotone: the closer the intra-dimension weights move to raw elicited judgment, the worse the ranking. Rendered from `results/ahp_shrinkage_sweep_v3.json`.

S4.1. Global Sensitivity of the Composite Oracle I_{comp} Severity Weights

The multi-metric composite failure oracle

$$I_{\text{comp}}(v) = w_1 \cdot \text{ReachabilityLoss} + w_2 \cdot \text{Fragmentation} + w_3 \cdot \text{ThroughputLoss} + w_4 \cdot \text{FlowDisruption} \quad (5)$$

defines failure impact across four operational dimensions, parameterized by the AHP-derived severity weights (0.35, 0.25, 0.25, 0.15). It is a secondary oracle that the main manuscript does not use: it labels the explanation layer’s evaluation in this supplement (Section S9) and enters the inter-oracle comparison of Section S12.

To determine whether this phenomenon is an artifact of the specific AHP weight vector or a structural property of type-heterogeneous aggregation, we conducted two global sensitivity analyses across the 4-simplex ($\sum_{i=1}^4 w_i = 1, w_i \geq 0$) using the manuscript’s twelve-scenario corpus (`results/icompsensitivity_jss12.json`):

1. **One-Factor-At-A-Time (OFAT) Sweeps:** Each weight w_i is swept across $[0.05, 0.70]$ in 14 increments, proportionally renormalizing the remaining three weights.
2. **Uniform Simplex Sampling:** $N = 1,000$ weight vectors sampled uniformly on the 4-simplex via $\text{Dirichlet}(1, 1, 1, 1)$, evaluating stratified and pooled Spearman ρ across all scenarios.
3. **Morris Elementary-Effects Screening:** Trajectory screening ($r = 15, p = 4$ levels) to isolate the influence of each severity weight on the stratum-separation gap $\Delta = \min(\rho_{\text{app}}, \rho_{\text{broker}}, \rho_{\text{node}}) - \rho_{\text{pooled}}$.

Table S4: Global sensitivity of $I_{\text{comp}}(v)$ severity weights across the 4-simplex ($N = 1,000$ Dirichlet draws and Morris elementary-effects screening). Shipped weights: Reachability 0.35, Fragmentation 0.25, Throughput 0.25, Flow Disruption 0.15. Measured over the manuscript’s twelve-scenario corpus. **Sweep Baseline** is the sweep harness’s own evaluation at those weights; it agrees with the canonical detection-benchmark figures quoted in Section 7.3.3 to within 0.002 on every stratum (see the reconciliation note below).

Stratum / Factor	Sweep Baseline	Dirichlet Simplex Range	Morris μ_{gap}^*	Morris σ_{gap}
Application ρ	0.597	[0.441, 0.599]	—	—
Broker ρ	0.317	[0.231, 0.427]	—	—
Node ρ	0.140	[0.107, 0.207]	—	—
Pooled ρ	0.218	[0.096, 0.351]	—	—
Reachability (w_1)	0.35	[0.05, 0.70]	0.174	0.094
Fragmentation (w_2)	0.25	[0.05, 0.70]	0.002	0.006
Throughput (w_3)	0.25	[0.05, 0.70]	0.231	0.075
Flow Disruption (w_4)	0.15	[0.05, 0.70]	0.143	0.113

Table S4 establishes three key findings:

1. **Stratum Dominance is Invariant:** Across all 1,000 Dirichlet draws, Application correlation ($\rho \in [0.441, 0.599]$) strictly dominates pooled correlation ($\rho \in [0.096, 0.351]$); the two intervals do not overlap. Heterogeneous pooling degrades ranking performance under 100% of sampled weight configurations, demonstrating that the failure of pooled evaluations is an intrinsic property of mixing distinct architectural strata rather than an artifact of weight tuning. This is the claim the stratification argument rests on, and it is the one that survives re-scoping: it held on the eight-scenario suite reported previously and holds on the twelve-scenario corpus reported here.
2. **The Strict Condition Does Not Hold at the Shipped Weights:** $\rho_{\text{pooled}} < \min(\rho_{\text{app}}, \rho_{\text{broker}}, \rho_{\text{node}})$ holds across 13.9% of the Dirichlet simplex, and the shipped weight vector is not among them: at $(0.35, 0.25, 0.25, 0.15)$ the gap is -0.078 , because pooled correlation (0.218) exceeds the Node stratum (0.140). On an eight-scenario suite belonging to a companion study, the condition holds at the shipped weights on 34.8% of the simplex, but that suite’s Broker stratum ($\rho = 0.119$ over six scenarios) is far weaker than this corpus’s (0.317 over eleven). The strict reversal is a property of that suite, not of type-heterogeneous aggregation.
3. **Dominant Drivers of Separation:** Morris screening demonstrates that Throughput ($\mu^* = 0.231, \sigma = 0.075$) and Reachability ($\mu^* = 0.174, \sigma = 0.094$) are the primary drivers of the stratum gap, with Flow Disruption third ($\mu^* = 0.143$), whereas Fragmentation ($\mu^* = 0.002$) exerts virtually no influence. On the eight-scenario suite Flow Disruption ranked first; the re-ordering is a further consequence of that suite’s atypically weak Broker stratum.

Reconciliation with Section 7.3.3. The *Sweep Baseline* column and the canonical detection-benchmark figures agree, and both are now computed over the same twelve scenarios the rest of the manuscript evaluates. The canonical values come from `results/detection_validation_jss12.json`, produced by running the

full **FailureSimulator**: $\rho = 0.597$ (Application), 0.317 (Broker), 0.138 (Node), 0.217 (pooled); the sweep, which evaluates I_{comp} as a direct linear combination of four cached per-node severity components and does not re-execute the simulator, reproduces each to within 0.002. Two corrections are recorded here. First, this note once described a discrepancy of up to 0.035 between the two procedures and attributed it to that methodological difference; the gap was in fact a defect in the simulator-side evaluation, and correcting it closed the gap on every stratum. Second, and more consequentially, both procedures previously ran over an eight-scenario suite belonging to a companion study, which omits five scenarios of this corpus and includes a regression fixture that appears in no other table of this manuscript. Every figure in this section is now measured over the manuscript’s own twelve, and the scenario list is shared by the two scripts rather than duplicated in each.

S5. Zero-Inflation Sensitivity of the Agreement Figures

Spearman ρ over a population where many components are tied at exactly zero is driven substantially by how those ties are handled, and $I^*(v)$ is heavily zero-inflated: 19 to 106 Applications per scenario carry exactly zero cascade impact, whereas $I_{\text{comp}}(v)$ has no exact zeros at all. We therefore report, alongside each full-population figure, $\rho_{>0}$ — the same correlation restricted to components both oracles score strictly positive. It is a sensitivity bound, not a replacement: for a component the simulator actually injected, zero impact is a real measurement ("its failure reaches nobody"), and dropping it would be a results-favorable filter.

The bound changes the reading of two scenarios, in opposite directions. On **hub_and_spoke**, I_{comp} and I^* correlate at $\rho = -0.044$ over the full population but $\rho_{>0} = +0.263$ over the 22 components both score positive: the apparent *sign* disagreement is an artifact of tied zeros, not evidence that the oracles order active components inversely. On **microservices**, the movement runs the other way — $\rho = 0.461$ falls to $\rho_{>0} = 0.257$, so a substantial part of that scenario’s agreement is the two oracles concurring on which components are inert rather than on how the active ones rank. Averaged over the seven scenarios the two summaries are close ($\rho = 0.425$, $\rho_{>0} = 0.447$), which is precisely why the per-scenario figures matter: the mean conceals compensating movements of 0.2–0.3 in both directions.

S6. Domain-Specific Weighting and Threshold Sensitivity

Sweeping the composite reliability weight $w_R \in [0, 1]$ moves mean ρ by only 0.018 (0.317 at $w_R = 0$ to 0.336 at $w_R = 1$), and the domain-derived and static rankings agree at mean Kendall $\tau = 0.980$ — the two orderings are very nearly the same ordering. Within that narrow band the domain-derived weighting does not improve on either alternative it replaces: mean ρ is 0.318 (domain-derived) against 0.331 (equal) and 0.321 (static), so it is marginally the *worst* of the three, losing to equal weighting by 0.013 and to static by 0.003. Both margins are far smaller than the between-model differences of Section 6.1 of the main manuscript and we draw no ranking conclusion from either. What the sweep establishes is that no weighting confined to this parameter could move ρ appreciably, so ISO/IEC 25019 context-of-use reweighting must be understood as an attributional device that expresses criticality in stakeholder terms — and not, on this evidence, as a mechanism that makes the ranking more accurate. Two free parameters of the simulated labels themselves matter more than any scoring weight, and we sweep both. The cascade propagation threshold moves mean ρ across a spread of 0.084 (0.189 at a threshold of 0, rising to 0.271 at 0.5 and flat thereafter at 0.273): a permissive threshold that lets every edge propagate produces a noisier target than one that requires meaningful coupling, and the ordering stabilizes once the cutoff reaches 0.5. Feature normalization spans 0.037 — robust scaling at 0.234 against 0.197 for both min–max and z -score. That is smaller than the threshold spread but of the same order, not negligible beside it: rank-based robust scaling and the two magnitude-preserving alternatives do not induce the same ordering, and the choice is a reported parameter of the scorer rather than an implementation detail. Neither spread indicates that the ordering is *correct*; they indicate only that it is stable under the free parameters of the oracle and the scorer, which is the weaker property a sensitivity sweep can establish.

S7. AHP Pairwise-Comparison Matrices and Their Consistency

Several weight vectors in this framework are derived by the Analytic Hierarchy Process. We state the two that back reported quantities, and we report a property of the matrix family that bears on how much the accompanying consistency ratios are worth.

A caveat on consistency ratios. Saaty’s consistency ratio detects *inconsistent* judgment; it does not detect a matrix constructed from its own answer. If a priority vector w is chosen first and entry (i, j) is then filled in as w_i/w_j , the resulting matrix is perfectly consistent, $CR \approx 0$, and the statistic certifies nothing. Such a matrix is rank-one: every row is a scalar multiple of every other. Writing $\sigma_1 \geq \sigma_2$ for its two largest singular values, the ratio σ_2/σ_1 separates the two cases — it is near zero for a back-filled matrix and appreciably positive for one carrying genuine judgment disagreement.

Table S5: Consistency diagnostics for the framework’s five AHP matrices. CR alone would suggest all five are sound; σ_2/σ_1 shows that three encode a declared priority vector rather than independent pairwise elicitation. The Availability matrix returns a slightly negative CR , which is not attainable for a genuinely elicited matrix ($\lambda_{\max} \geq n$) and arises here as floating-point noise around exact consistency.

Matrix	n	CR	σ_2/σ_1
Topic QoS (main manuscript, Eq. 3)	3	+0.0158	0.072
Fault Tolerance	3	+0.0029	0.027
Composite impact (I_{comp})	4	+0.0011	0.014
Maintainability	5	+0.0005	0.0006
Availability	5	−0.0029	0.0006

Only the Topic QoS matrix carries a consistency ratio that means what a consistency ratio is supposed to mean, and it is the one matrix a continuous-integration test guards on both sides: `tests/test_ahp_shrinkage.py` asserts $0.005 < CR < 0.10$, the lower bound existing precisely as a non-degeneracy check. We report the remaining vectors as declared constants with documented structure rather than as elicited judgment, and note that this reading costs the framework little: the global sensitivity analysis (Section S4) finds eight of the ten weight constants to have $\mu^* \leq 0.023$, so the ranking results in Section 6 of the main manuscript do not rest on these values being optimal.

Table S6: Saaty matrix over the three Topic QoS dimensions backing $QoS(t)$ in Equation (3) of the main manuscript. Geometric-mean priority vector (0.2385, 0.6250, 0.1365); $\lambda_{\max} = 3.018$, $CI = 0.0092$, $RI = 0.58$, $CR = 0.0158$. The shipped constants (0.24, 0.62, 0.14) round this within 0.005.

	Reliability	Durability	Priority
Reliability	1	1/3	2
Durability	3	1	4
Priority	1/2	1/4	1

Table S7: Matrix over the four composite-impact criteria — reachability loss (RL), fragmentation (FR), throughput loss (TL), flow disruption (FD) — backing $I_{\text{comp}}(v)$ (Section S4.1). Raw priority vector (0.389, 0.255, 0.255, 0.100); after the framework’s $\lambda = 0.7$ shrinkage toward a uniform prior, (0.347, 0.254, 0.254, 0.145), which the shipped (0.35, 0.25, 0.25, 0.15) round. Per Table S5 this matrix is rank-one, so it documents the provenance of those constants without independently justifying them.

	RL	FR	TL	FD
RL	1	3/2	3/2	4
FR	2/3	1	1	5/2
TL	2/3	1	1	5/2
FD	1/4	2/5	2/5	1

Table S8: The Reliability–Maintainability (RM) quality decomposition (supporting Section S27).

Dimension	Sub-Characteristic	Architectural Question	Underlying Graph Metrics	Role / Remediation
Reliability (<i>R</i>)	Fault Tolerance (<i>FT</i>)	How broadly does failure propagate?	Reverse PageRank on G^T , in-degree, cascade depth	Reliability Eng.: add redundancy, circuit breakers
	Availability (<i>A</i>)	Is this a single point of failure?	Directed articulation score (raw + QoS-weighted), bridge ratio, CDI	DevOps/SRE: replicate host/broker
Maintainability (<i>M</i>)	Modularity/Modifiability	How complex and coupled is this?	Betweenness, QoS-weighted out-degree, Code Penalty, clustering	Architect: refactor code, decouple

S8. Generative Parameters of the Synthetic Corpus

Table S9 records the configuration behind each synthetic scenario of Table 3 in the main manuscript. It is replication detail rather than a result: the corpus shape a reader needs to interpret the evaluation is in that table itself, and every configuration is committed in the replication package.

Table S9: Generative parameters of the twelve synthetic scenarios (eleven evaluation scenarios plus the ATM case study). Seed and fan-out figures are read from the committed configurations; per-entity counts are in Table 3 of the main manuscript. The modal QoS column gives the most common reliability/durability/priority value and the range of topic shares carrying them, computed from the committed topology rather than the config’s declared QoS targets, which domain-driven assignment does not always realize (Section 5.1 of the main manuscript).

Scenario	Seed	Pub	Sub	Modal QoS (R/D/P)
Air Traffic Management (ATM)	42	1.2	2.9	RELIABLE/VOLATILE/HIGH (41–81%)
Autonomous Vehicle (AV)	1001	2.5	5.0	RELIABLE/TRANSIENT_LOCAL/HIGH (45–80%)
Enterprise Pub-Sub	7007	3.0	4.5	RELIABLE/TRANSIENT_LOCAL/MEDIUM (29–79%)
Financial Trading	3003	4.0	6.0	RELIABLE/PERSISTENT/HIGH (40–89%)
Healthcare Integration	4004	2.5	3.0	RELIABLE/PERSISTENT/HIGH (40–88%)
Enterprise Integration (ESB)	5005	2.0	7.0	RELIABLE/TRANSIENT_LOCAL/MEDIUM (33–67%)
Industrial SCADA	1902	1.1	1.0	RELIABLE/TRANSIENT/HIGH (31–80%)
IoT Smart City	2002	2.0	1.5	BEST_EFFORT/VOLATILE/LOW (56–75%)
Logistics Fleet	2104	1.8	1.9	RELIABLE/TRANSIENT_LOCAL/MEDIUM (40–76%)
Microservices Mesh	6006	1.5	2.0	RELIABLE/TRANSIENT_LOCAL/MEDIUM (40–60%)
Real-Time Gaming	2003	2.6	2.8	BEST_EFFORT/VOLATILE/HIGH (39–71%)
Telecom RAN	1801	2.2	1.6	RELIABLE/VOLATILE/HIGH (36–67%)

S9. Anti-Pattern Detection and Node-Type Stratification

Validated against $I_{\text{comp}}(v)$, the rule-based anti-pattern catalog reaches mean precision 0.244 and recall 0.900 ($F_1 = 0.384$) — but it implicates 93.4% of scored components, so that recall is close to what indiscriminate flagging achieves and the precision is near the base rate. The ATM scaling sweep (29 to 444 components, five seeds, seed-invariant) shows the mechanism: Cohen’s κ ends at -0.018 on the 444-component instance, against $+0.035$ at 29 components, with a flag rate of 92.3% at the largest scale. The path is not monotone — κ rises to $+0.086$ at 148 components before collapsing — so we do not claim a clean decay law, only the endpoint: at the largest scale the catalog’s agreement with the oracle is indistinguishable from chance, and marginally on the wrong side of it. The catalog does not become wrong at scale so much as it stops discriminating. We report it as a characterization of the catalog rather than a working triage mechanism, and delegate critical-set identification to the continuous rankers of Sections 6.1–6.2 of the main manuscript. A nineteenth detector, DEEP_PIPELINE, is excluded for path-combinatorial explosion (247,761 paths on a 29-component fixture).

Stratification vs. Pooling: Measured against the composite oracle $I_{\text{comp}}(v)$ over the twelve scenarios of the corpus, stratified RM rank correlations are $\rho = 0.597$ (Application, 12 scenarios), 0.317 (Broker, 11 scenarios), and 0.138 (Execution Host, 12 scenarios), while pooled correlation across all types is $\rho = 0.217$. Pooling node types conflates disparate structural base rates across architectural layers, leaving the pooled figure at less than two fifths of the Application stratum. It is not a strict Simpson reversal on this corpus: pooled correlation sits above the Node stratum (0.138). This is why every evaluation in this paper is reported on a single stratum, and why pooled critical-set numbers should be read as inflated wherever they appear

(Section 6.3 of the main manuscript). On that same pooled benchmark, unweighted degree centrality attains $\rho = 0.314$ and $F_1 = 0.459$ against RM’s 0.217 and 0.350 — RM is beaten by degree on both, confirming that it serves as an attribution instrument rather than an unconstrained global ranker.

S10. Explanation Layer on the Open-Source System Models

We evaluated the framework on hand-authored architecture models of five open-source systems (Section 5.1 of the main manuscript), written as dedicated adapters from the systems’ public documentation, with results presented in Table S10. We note the scope of this evaluation before reporting it. Online Boutique is a vendor demonstration application and Train-Ticket an academic benchmark, Home Assistant is an authentic open-source IoT automation platform, and EdgeX Foundry is an industrial edge computing reference implementation; each is an abstraction of its upstream repository rather than a complete transcription; and their labels come from the same simulation oracle used throughout, so what follows tests topological generalization, not agreement with observed field failures. Two further limits bound what this table can support. First, the predictor scored in Table S10 is the deterministic explanation layer $Q(v)$ together with the two training-free topological baselines; the learned model is evaluated separately in Section 6.3 of the main manuscript. Both are scored against $I^*(v)$, but under different injector settings (see the caption), so the two tables are not commensurable. Second, the labels are strongly tied at zero in several architectures — 13 of 32 Applications in Autoware, 14 of 22 in Cloud Microservices, 27 of 41 in Train-Ticket, and 12 of 22 in EdgeX carry zero simulated impact (the complement of the “Impactful Apps” column), whereas Home Assistant exhibits much lower zero-inflation with 17 of 24 applications actively participating in failure cascades. Spearman ρ over populations that are heavily tied is dominated by how those ties are handled; the values below use midrank ties throughout.

Table S10: Open-source reference systems scored by the deterministic explanation layer RM / $Q(v)$ — no GNN checkpoint is invoked, so $\pm$ is oracle variance over five simulation seeds, not training variance. “Gain vs. Deg” is $\rho_Q - \rho_{\text{deg}}$. The labels here are $I^*(v)$ (cascade reachability injection) — the same oracle as Table 9 of the main manuscript, but run with the validation CLI’s cascade-depth cap of 5 and averaged over five injector repeats per node, where that table’s labels run to unlimited depth. Simulator-derived labels, not observed incidents.

System model	$ V $	$ V_{\text{app}} $	Spearman ρ	Kendall τ	App Overlap@K	Pooled Overlap@K	Impactful Apps	Gain vs. Deg
Autoware.universe (ROS 2)	75	32	0.685 $\pm$ 0.010	0.513	0.333	0.800	19 / 32	+0.357
Cloud Microservices Mesh	60	22	0.778 $\pm$ 0.001	0.639	0.500	1.000	8 / 22	+0.014
Train-Ticket Booking Mesh	90	41	0.759 $\pm$ 0.001	0.605	0.625	1.000	14 / 41	+0.264
Home Assistant (Smart Home)	63	24	0.514 $\pm$ 0.016	0.373	0.250	0.667	17 / 24	+0.289
EdgeX Foundry (Industrial IoT)	63	22	0.800 $\pm$ 0.015	0.563	0.000	1.000	10 / 22	+0.427

Key Insights for RQ4:

1. **The explanation layer ranks unseen architectures well, but this is not a learned-transfer result.** RM / $Q(v)$ attains $\rho = 0.800$ on EdgeX Foundry, 0.778 on Cloud Microservices, 0.759 on Train-Ticket, 0.685 on Autoware.universe, and 0.514 on Home Assistant. Because $Q(v)$ is a closed-form scoring function rather than a fitted model, applying it to a new architecture involves no transfer in the inductive sense of Section 6.2 of the main manuscript: nothing was trained, and no distribution shift is being survived. What this table establishes is that the deterministic attribution of Section S27 of the main manuscript remains informative on architectures we did not generate. The learned model is a separate question, answered in Section 6.3 of the main manuscript. Table S10 is not commensurable with RM’s mean LOSO correlation ($\rho = 0.205$, Table S34), but the cause is not the oracle: Table S34, Table S28 and Table S10 all score against $I^*(v)$ (cascade reachability injection). Table S34 is the synthetic LOSO corpus, whereas Table S10 and Table S28 both score the five system models on the Application stratum; under the unlimited-depth labels of Table S28 (the run behind Table 9 of the main manuscript) RM attains $\rho = 0.516$ on those same systems. What separates the figures is therefore a real-versus-generated difference together with the injector configuration recorded in the caption, with the tied-label structure described above contributing to both. The practical consequence is that no RM figure in this paper should be compared across tables without checking which population and which injector configuration produced it.

2. **Stratified vs. Pooled Critical Set Identification:** On the stratified Application population (V_{app} with $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$), $Q(v)$ identifies the top-20% critical services with $F_1@K = 0.333$ in Autoware, 0.500 in Cloud Microservices, 0.625 in Train-Ticket, 0.250 in Home Assistant, and 0.000 in EdgeX Foundry (where failure impact is dispersed broadly across active device services). When evaluated across the pooled multigraph ($K = \text{round}(0.20 \cdot |V|)$), the pooled $\text{Overlap}@K$ reaches 0.800, 1.000, 1.000, 0.667, and 1.000, respectively. This disparity is the base-rate phenomenon of Section 5.3 of the main manuscript: the passive infrastructure entities (Topics, Nodes, Libraries) carry constant zero failure impact under the injector configuration used for these runs, so a pooled top- K set is largely populated by correctly identifying inert nodes. The pooled column is therefore inflated and we do not read it; the stratified V_{app} column is the operational figure.
3. **Framework Acceptance Gates and Cross-Domain Disparities:** The framework ships a four-condition release gate ($\rho \geq 0.75$, $\text{Overlap}@K \geq 0.70$, $\text{SPOF } F_1 \geq 0.60$, prediction gain ≥ 0.02). EdgeX Foundry passes all four gates across all five seeds (100% pass rate), driven by strong rank correlation ($\rho = 0.800$), pooled $F_1 = 1.00$, and robust articulation detection ($\text{SPOF } F_1 = 0.80$). In contrast, the pass rate is zero on the other four architectures: Autoware fails the ρ and SPOF conditions, Cloud Microservices fails SPOF and prediction gain, Train-Ticket fails SPOF , and Home Assistant misses the ρ gate ($\rho = 0.514$) despite scoring a perfect $\text{SPOF } F_1 = 1.00$. What this establishes is a negative result about thresholds, not a positive one about ordering: a gate calibrated on our synthetic corpus fires correctly on one architecture in five, so its absolute cut-offs are domain-specific and do not transfer as shipped. The ordering half of the question — whether the ranking underneath those thresholds transfers — is not answered here and is answered negatively in Section 6.3 of the main manuscript, once the population is restricted to components that actually propagate a failure.
4. **Predictive Advantage over Structural Heuristics:** The “Gain vs. Deg” column reports $Q(v)$ ’s rank-correlation margin over an unweighted degree centrality baseline (+0.427 on EdgeX Foundry, +0.357 on Autoware, +0.289 on Home Assistant, +0.264 on Train-Ticket, +0.014 on Cloud Microservices), so on this population and this oracle the hierarchical attribution of Section S27 of the main manuscript orders components better than degree centrality does. The margin does not generalize beyond those conditions, and we scope it accordingly: on the pooled synthetic detection benchmark of Section 5.3 of the main manuscript — which is measured against $I_{\text{comp}}(v)$ (composite failure simulation) rather than the $I^*(v)$ used here, and across all entity types — degree *beats* RM on both rank correlation (0.314 vs. 0.217) and $\text{Overlap}@K$ (0.459 vs. 0.350). The two findings are not in conflict — one is stratified on Applications in five hand-authored system models under $I^*(v)$, the other pooled across types in eight generated ones under $I_{\text{comp}}(v)$ — but neither supports a general claim that RM dominates degree. Wilcoxon signed-rank tests over the five systems do not reach significance ($p \geq 0.33$), which at $n = 5$ they could hardly do; we report them for completeness rather than as support.
5. **The QoS-weighted heuristic was initially unscorable here, and the fix is the substrate.** On the raw multigraph the unweighted Topo reference is computable ($\rho = 0.307/0.891/0.528$ on Autoware, Online Boutique and Train-Ticket; mean 0.511) while Topo-QoS is not. The obstacle is not missing QoS data — every topic in all five adapters carries declared **durability**, **reliability** and **transport_priority** profiles, and projecting $w(t)$ onto the structural edges yields non-unit weights on 45.3–62.2% of them. It is that Topo reads betweenness off the cached **DEPENDS_ON** projection, whereas the QoS-weighted variant must recompute it on the raw multigraph, where Applications never route messages and betweenness is identically zero for all of them — the same degeneracy Section 5.2 of the main manuscript gives as the reason topological baselines are run on the projection at all. This is resolved in the reported evaluation: both baselines are scored on the **DEPENDS_ON** flow projection, as Section 5.2 of the main manuscript prescribes, and Table 9 of the main manuscript therefore carries Topo-QoS on all five systems (0.289–0.888). Separately, projection-based baselines assign constant zero to Applications carrying no derived **DEPENDS_ON** edge, accentuating sensitivity to tied labels.

S11. HGT Relational Attention Weight Analysis

Figure S2 visualizes the relational mutual attention distributions of a Heterogeneous Graph Transformer on the ATM case study architecture. *The model is not the predictor evaluated elsewhere in this paper.* It is a three-layer, four-head HGT trained for 100 epochs on the ATM scenario alone at seed 42, built by `reproduce/extract_attention.py` for this figure and discarded afterwards. The LOSO-evaluated checkpoints cannot stand in: they are trained over the full corpus, carry a 40-relation schema against this topology’s 32, include `Library` publish and subscribe relations the ATM graph does not contain, and are two layers deep rather than three. Nothing below therefore speaks to what the evaluated predictor attends to; it speaks to whether typed attention differentiates at all.

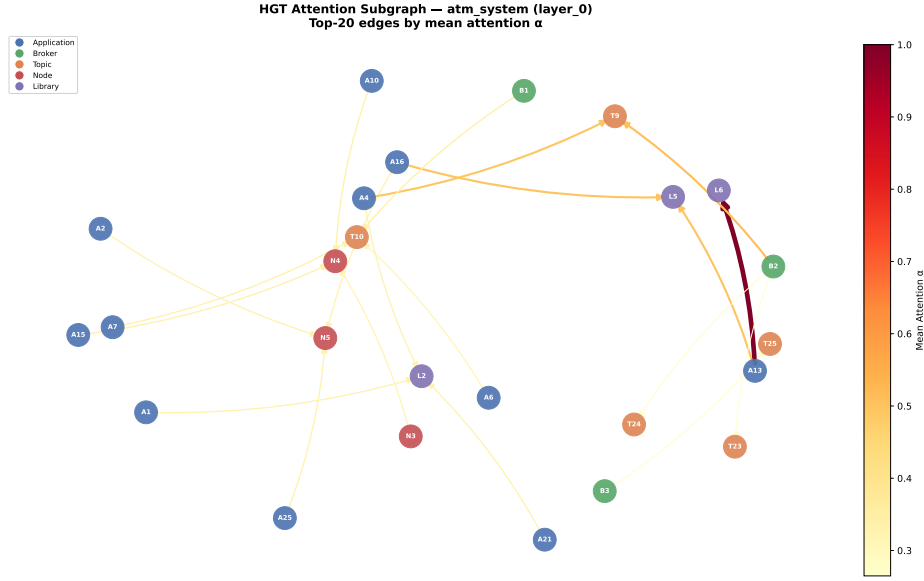


Figure S2: Relational attention on the ATM case study, from a three-layer HGT trained for 100 epochs on this scenario alone (seed 42) — not the predictor evaluated in Sections 6.1–6.3 of the main manuscript. Mean α spans 0.113–0.222 across the eight relation types and is governed substantially by destination in-degree. A qualitative illustration on one topology, not evidence of general relation prioritization.

Aggregated by relation type, first-layer mean attention across all eight relations runs: `USES` (App→Lib) 0.220; `ROUTES` 0.187; `SUBSCRIBES_TO` 0.172; `USES` (Lib→Lib) 0.169; `PUBLISHES_TO` 0.160; `RUNS_ON` (App→Node) 0.156; `CONNECTS_TO` 0.135; and `RUNS_ON` (Broker→Node) 0.113. Across the three layers no relation-type mean moves by more than 0.019, but the rank order is *not* stable: the four relations between 0.15 and 0.18 are spaced more closely than that drift, and they permute between layers. Three methodological cautions bound what this visualization supports:

1. The spread across all eight relation types is narrow (0.113–0.222, and only 0.156–0.220 among the five carrying more than nine edges), indicating a mild distributional tendency rather than a clear-cut structural separation.
2. Because α is computed as a softmax over each destination node’s incoming neighborhood $\mathcal{N}(v)$, the mean attention weight α is substantially governed by local in-degree. The single largest weight in the whole extraction ($\alpha_{uv} = 1.00$, on a `USES` edge into a library) lands on a destination of in-degree one, where $\alpha = 1.0$ holds by arithmetic and carries no information. The thinnest relations sit at the extremes of the ordering: `USES` (Lib→Lib) carries 3 edges and `RUNS_ON` (Broker→Node) carries 5, against 85 for `SUBSCRIBES_TO`.
3. The figure is a function of the corpus snapshot, not a stable property of the architecture. Regenerating the ATM topology changed its edge set without changing any entity count (`PUBLISHES_TO` 56 → 48,

SUBSCRIBES_TO 90 $\rightarrow$ 85, ROUTES 36 $\rightarrow$ 37, CONNECTS_TO 7 $\rightarrow$ 9), and that alone moved USES (Lib $\rightarrow$ Lib) from 0.225 to 0.169 — from above ROUTES to below it. The extraction itself is deterministic, so the whole of that shift is input sensitivity on a 74-node graph.

Consequently, Figure S2 serves as an illustrative artifact confirming that multi-head heterogeneous attention remains active across relation types, rather than a statistical proof of general relational importance.

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S12. Convergent Validity Over Simulation Oracles

We evaluated inter-oracle agreement across $I^*(v)$ (cascade reachability injection), $I_{\text{comp}}(v)$ (composite failure simulation), and $I_{\text{dyn}}(v)$ (dynamic queue-flow simulation) over the twelve inductive folds of Table S34, summarized in Table S11:

Table S11: Inter-oracle agreement across simulation paradigms (chance top- K Jaccard is 0.111) over the twelve LOSO topologies, Application population. One comparison per topology: the five seeds enter I^* only, averaged into a single label per component, while I_{comp} and I_{dyn} each run once at seed 42. I_{dyn} denotes the queue-flow discrete-event simulation oracle, I^* denotes the graph topological cascade injection oracle, and I_{comp} denotes the multi-criteria composite oracle. ρ^+ restricts the correlation to components both oracles score non-zero, separating directional agreement from agreement on which components are harmless. I_{dyn} was measured under enforced QoS contracts at a calibrated per-subscriber operating point ($\rho_{\text{util}} = 0.65$); the run is reproducible from the commit its artifact names.

Oracle pair	Mean ρ (range)	Mean ρ^+	Mean τ	Jaccard@ K	Tie-robust
I_{dyn} vs. I^*	0.627 (0.186–0.953)	0.429	0.493	0.370	0.373
I_{comp} vs. I^*	0.395 (0.083–0.653)	0.353	0.290	0.266	0.261
I_{comp} vs. I_{dyn}	0.411 (−0.073–0.705)	0.389	0.290	0.324	0.324

The queue-flow simulator and the topological cascade oracle agree substantially but not interchangeably ($\rho = 0.627$, Jaccard 0.370 against 0.111 expected by chance). **This is the reading the ceiling supports.** I^* ’s own seed-to-seed test–retest across these same twelve folds runs 0.811–1.000 (median 0.982; Section 6.1 of the main manuscript), so I_{dyn} agrees with I^* distinctly *less* closely than I^* agrees with itself. The gap is the point: a behavioral oracle that reproduced the topological one to within label noise would be re-measuring

the topology rather than corroborating it. Ranking over discrete-event queueing traffic recovers most of the topological ordering while retaining content the cascade abstraction does not express.

Two boundaries qualify this. First, $\rho^+ = 0.429$ against $\rho = 0.627$ shows that a substantial share of the agreement is the two oracles concurring on which components are *harmless*; restricted to components both score non-zero, agreement is weak in four of twelve folds (Industrial SCADA -0.069 , Enterprise Integration (ESB) 0.008 , Autonomous Vehicle 0.149 , Microservices 0.194). Second, Industrial SCADA is the weakest fold ($\rho = 0.186$), and Microservices exhibits the least reproducible label in the corpus (test–retest 0.811 , top- K Jaccard 0.720), so part of that deficit is label noise rather than construct divergence.

Both oracles have a floor, and only one of them had been measured. The reading above is stated against I^* ’s test–retest alone, which silently treats I_{dyn} as noiseless. It is not: it is a stochastic discrete-event simulation run here at a single seed. Re-running both non-primary oracles across seeds $\{42, 123, 456\}$ on three folds spanning the corpus — ATM, Healthcare Integration, and Microservices Mesh, the last being the weakest fold above — gives I_{comp} a test–retest of 1.000 everywhere (its Bernoulli cascade draws are effectively deterministic at the shipped edge weights, so its single seed costs nothing), and I_{dyn} a test–retest of 0.958 – 0.972 on ATM, 0.828 – 0.867 on Healthcare, and 0.741 – 0.821 on Microservices. The subset is three folds rather than twelve because I_{dyn} costs one discrete-event run per component per seed; it is chosen to bracket the corpus rather than to average over it.

The consequence is specific. On Microservices, I_{dyn} agrees with *itself* at 0.741 – 0.821 — at or below I^* ’s own 0.811 on that same fold — so the $\rho = 0.337$ reported there is bounded by two dispersions, not one, and attributing the shortfall primarily to I^* ’s label noise overstates what can be separated. Elsewhere the 0.627 headline is less affected: on ATM and Healthcare I_{dyn} reproduces itself at 0.83 – 0.97 , well above the cross-oracle agreement, so the construct-divergence reading of the gap survives on those folds. We report the floors rather than correcting the headline for them, because a noise-corrected correlation would require the twelve-fold multi-seed run this subset stands in for.

Four Golden Signals Telemetry & Dynamic Contention Relief. In addition to the scalar delivery rate drop $I_{\text{dyn}}(v)$, each discrete-event execution logs Google SRE’s Four Golden Signals (latency, traffic, errors, saturation) windowed across pre-fault ($t < t_{\text{fault}}$) and post-fault ($t > t_{\text{fault}}$) intervals. The queue-level decomposition shows that end-to-end message traversal latency t_{e2e} is driven by buffer waiting delay under contention ($\rho_{\text{util}} = 0.65$), whereas CPU service processing remains stationary. When a high-throughput publisher fails, surviving subscriber buffers rapidly drain, yielding a pronounced negative correlation between delivery drop I_{dyn} and tail latency delta ($\rho = -0.499$) as well as deadline violations ($\rho = -0.418$). Preserving the Four Golden Signals as first-class, structured telemetry avoids an additive cancellation trap and equips operators with comprehensive diagnostics alongside the convergent-validity ranking probe.

The Reachability Oracle I^ in Pseudocode*

The procedure below is `FaultInjector.cascade` in `saag/simulation/fault_injector.py`, with its default parameters (propagation threshold $\tau = 0.2$, depth-damping step 0.15 , floor 0.25 , no depth limit). For one injected component v and one seed:

1. Set $F \leftarrow \{v\}$ (if v is a Host, add every component that runs on it) and the frontier to F .
2. Repeat for waves $k = 0, 1, \dots$ while the frontier is non-empty or a topic newly loses feed:
 - (a) *Library blast.* Every component with a `DEPENDS_ON` edge to a failed Library joins F and the next frontier.
 - (b) *Topic loss.* For each topic t , $L(t) = \max(\text{rate share of failed publishers, share of failed routing brokers})$, multiplied by the QoS ladder factor ($\times 1.2$ for `RELIABLE`; $\times 1.15$ for high and $\times 1.05$ for medium transport priority) and clamped to $[0, 1]$; a failed topic has $L(t) = 1$.
 - (c) *Subscriber failure.* For each live subscriber s (in identifier order), $\ell(s)$ is the mean of $L(t)$ over its feeds. If $\ell(s) \geq \tau$, s fails with probability $\min(1, \ell(s)/\tau) \cdot \max(0.25, 1 - 0.15k)$, drawn from the seeded generator; failed subscribers join F and the next frontier.
3. Recompute $L(t)$ and $\ell(s)$ for the final F ; the seed’s impact is the mean of $\ell(s)$ over all subscribers.

$I^*(v)$ is the mean of the per-seed impacts over seeds $\{42, 123, 456, 789, 2024\}$. In the first wave the failure probability is one whenever $\ell(s) \geq \tau$, so the seeds act only on later waves. Subscribers, publishers and feeds are iterated in sorted order so that the result does not depend on the Python hash seed.

S13. In-Distribution Significance Tests

Paired tests over the twelve in-distribution scenarios of Table S18. They are reported here rather than in the body because Section S35 declines to read the in-distribution typed–untyped contrast as evidence about typing: the typed pair consumes the native multigraph while the homogeneous pair consumes the Application–Library projection, so the margin mixes typed message passing with multi-entity visibility. The tests are given for completeness, not as support for a claim.

Table S12: Paired Wilcoxon signed-rank tests across in-distribution scenarios ($n = 12$, two-sided).

Comparison	$\Delta\rho$	Won	Wilcoxon W	p -value	Significance
HGT-QoS vs. Topo	+0.291	11/12	2.0	0.0015	Significant ($p < 0.01$)
Topo-QoS vs. Topo	+0.198	12/12	0.0	0.0005	Significant ($p < 0.001$)
HGT-QoS vs. GAT-S-P-w	+0.222	11/12	3.0	0.0024	Significant ($p < 0.01$)
HGT-QoS vs. GAT-S-P	+0.118	9/12	21.0	0.1763	Not significant
HGT-QoS vs. Topo-QoS	+0.093	9/12	23.0	0.2334	Not significant
HGT vs. GAT-S-P	+0.082	8/12	29.0	0.4697	Not significant
HGT-QoS vs. HGT	+0.037	8/12	32.0	0.6221	Not significant
GAT-S-P-w vs. Topo-QoS	−0.129	4/12	21.0	0.1763	Not significant

S14. Typed Node Feature Schema

Both pathways read the same typed node properties from G_{analysis} : the ranking pathway (Section 4 of the main manuscript) projects them per entity type before heterogeneous message passing, and the explanation layer (Section S27) aggregates them into its quality profile. All five entity types share indices 0–17, an 18-dimensional base block of deterministic topological metrics produced by the static analysis stage whose execution cost is characterized in Section 6.4 of the main manuscript. Table S13 lists the complete 18-dimensional base schema.

Type-specific features extend that block:

- **Application (23 dims):** indices 18–22 add source-code metrics from SCA — lines of code, cyclomatic complexity, Martin’s instability $I_{\text{code}} = C_e/(C_a + C_e)$ [3], Lack of Cohesion in Methods, and the composite Code Quality Penalty (CQP).
- **Library (25 dims):** the Application block plus two library-specific blast-radius drivers (23–24): the normalized size of the transitive reverse-USES closure, and the normalized count of distinct subscribers reachable from topics published within that closure.
- **Broker (19 dims):** index 18 is normalized queue buffer capacity.
- **Topic (22 dims):** indices 18–21 are publisher count, subscriber count, log message frequency $\log(1 + \text{freq})$, and ordinal QoS criticality.
- **Infrastructure Node (20 dims):** indices 18–19 are normalized CPU core allocation and physical memory.

S15. Corpus Subset Behind Each Analysis

Not every analysis in Section 6 of the main manuscript runs on the full corpus: the sensitivity sweeps and the detection benchmark predate the four extended domains and operate on smaller cached subsets. This table states which subset backs each reported figure.

Table S13: The 18-dimensional base topological feature schema (indices 0–17) shared across all entity types, computed on G_{analysis} . Every metric is normalized to $[0, 1]$ within its graph.

Index	Symbol	Metric Name	Description and Architectural Role
0	$PR(v)$	PageRank	Downstream authority under random walk with restart ($\alpha = 0.85$).
1	$RPR(v)$	Reverse PageRank	Upstream dependency exposure (transposed graph random walk).
2	$BT(v)$	Betweenness Centrality	Fraction of all-pairs shortest dependency paths traversing entity v .
3	$CL(v)$	Closeness Centrality	Reciprocal of average shortest-path distance to all reachable entities.
4	$EV(v)$	Eigenvector Centrality	Influence score weighted by principal eigenvector of adjacency matrix.
5	$DG_{\text{in}}(v)$	In-Degree Centrality	Normalized count of incoming dependency edges ($\deg^-(v)/(V - 1)$).
6	$DG_{\text{out}}(v)$	Out-Degree Centrality	Normalized count of outgoing dependency edges ($\deg^+(v)/(V - 1)$).
7	$CC(v)$	Clustering Coefficient	Local transitivity and interconnectivity among neighboring entities.
8	$AP(v)$	Undirected Articulation	Biconnected component cut-vertex indicator (binary cut score).
9	$BR(v)$	Bridge Ratio	Fraction of incident edges that constitute structural bridges.
10	$w(v)$	Node QoS Weight	Normalized aggregate criticality of incident transport contracts.
11	$w_{\text{in}}(v)$	QoS In-Degree	Sum of QoS coupling weights on incoming dependencies ($\sum_u w(u, v)$).
12	$w_{\text{out}}(v)$	QoS Out-Degree	Sum of QoS coupling weights on outgoing dependencies ($\sum_u w(v, u)$).
13	$MPCI(v)$	Multi-Path Coupling	Afferent multi-topic channel density across alternative paths.
14	$PC(v)$	Path Complexity	Mean efferent channel multiplicity: $\frac{1}{ \text{Out}(v) } \sum_e \log_2(1 + \text{path_count}(e))$.
15	$FOC(v)$	Fan-Out Criticality	Topic-modulated subscriber blast radius (0.0 for non-topic entities).
16	$AP_c^{\text{dir}}(v)$	Directed Articulation	Strongly connected component cut-vertex indicator.
17	$CDI(v)$	Connectivity Degradation	Change in all-pairs reachability when entity v is removed.

Table S14: Which corpus subset backs each analysis. Figures from different rows are not directly comparable, and we do not compare them.

Analysis	Scenario subset	n	Reported in
In-distribution ranking	Eleven evaluation scenarios + ATM	12	Tables S18, S12
Inductive LOSO	Eleven evaluation scenarios + ATM	12	Table S34, main Sections 6.1–6.2
QoS edge-feature ablation	Same twelve LOSO folds	12	Main Section 6.2
Weight sweeps and Morris screening	Six–seven core synthetic domains	6–7	Supplementary S1
Cross-oracle convergent validity	Same twelve LOSO folds	12	Main Section 4.3, Section S12
Anti-pattern detection, stratification	Seven core domains + ATM	8	Main Section 5.3, Section S9
Relational attention illustration	ATM case study alone	1	Supplementary S8
Real-world zero-shot transfer	Five open-source systems	5	Main Table 9, Section S10

S16. Per-Scenario Corpus Composition

Entity and edge counts for every scenario, read from the committed topology files rather than from the generator configurations. Continuous integration asserts that each dataset regenerates byte-identically from its configuration and that these counts match what is on disk.

How the two RPC systems were modeled. The five open-source system models are encoded by hand in `saag/adapters/realworld_adapter.py`; no importer reads launch files or deployment manifests. Online Boutique and Train-Ticket communicate by synchronous RPC (gRPC and REST, respectively), and their models re-express them as event-driven publish-subscribe meshes rather than decomposing each call into a request and a reply. The Online Boutique model has 22 Applications and 20 event topics on four brokers (Kafka, RabbitMQ, Redis and NATS), none of which the reference deployment uses; the original has about eleven gRPC services and no broker. The Train-Ticket model has 41 Applications and 30 topics on three brokers, one of which is the Eureka service-discovery server modeled as a broker; three topics carry requests or commands and none carries replies. Neither model therefore keeps request-reply coupling, timeouts, thread-pool exhaustion or synchronous backpressure, and failures propagate in them only along the declared publish-subscribe paths.

Table S15: Experimental evaluation corpus. The twelve synthetic topologies are the inductive Leave-One-Scenario-Out folds of Table S34; the five real-world systems are withheld from every training fold and used only for zero-shot transfer (Section 6.3 of the main manuscript). Counts are read from the committed topology files rather than from the generator configurations.

Dataset / Architecture	System Paradigm	V	V _{app}	Topics	Brokers	Hosts	Libs	E
<i>Synthetic evaluation scenarios (evaluation role in the manifest)</i>								
Autonomous Vehicle (AV)	ROS 2 Cyber-Physical	152	80	40	4	8	20	774
Enterprise Pub-Sub	Kafka Event Mesh	520	300	120	10	40	50	3,216
Financial Trading	Low-Latency Pub-Sub	124	60	35	5	6	18	631
Healthcare Integration	HL7/FHIR Event Mesh	98	50	25	3	8	12	389
Enterprise Integration (ESB)	Enterprise Application Integration (Broker Hub)	139	70	30	2	12	25	691
IoT Smart City	MQTT Telemetry Mesh	326	200	80	6	30	10	1,188
Microservices Mesh	Cloud-Native Services	186	90	45	6	15	30	678
Telecom RAN	5G Radio Access Network	225	120	55	8	20	22	881
Industrial SCADA	Plant Control Telemetry	254	140	70	4	25	15	824
Real-Time Gaming	Multiplayer State Sync	158	75	38	5	12	28	630
Logistics Fleet	Vehicle Telematics Mesh	205	110	50	7	18	20	755
<i>Synthetic case study (LOSO fold; case_study role in the manifest)</i>								
Air Traffic Management (ATM)	ICAO Global ATM Concept	74	26	27	5	8	8	261
<i>Hand-authored open-source system models (never used as training folds)</i>								
Autoware.universe [4]	Model of ROS 2 Autoware	75	32	24	3	6	10	179
Cloud Microservices [5]	Pub-sub model after GCP Online Boutique	60	22	20	4	6	8	128
Train-Ticket [6]	Model of Train-Ticket booking	90	41	30	3	8	8	162
Home Assistant [7]	Model of Smart Home	63	24	22	3	6	8	119
EdgeX Foundry [8]	Model of Industrial IoT	63	22	24	3	6	8	112
Synthetic subtotal (12 LOSO folds)		2,461	1,321	615	65	202	258	10,918
Open-source system models (5)		351	141	120	16	32	42	700
Total		2,812						11,618

S17. Running Example: Structural Graph and Its Projection

The running example now appears in the main manuscript as Figure 2, next to the projection rules it illustrates (Section 3.2 of the main manuscript). It is generated by `reproduce/render_jss_diagrams.py`.

S18. Identification Quality Beyond Top- K Overlap

Every $\text{Overlap}@K$ figure in the main manuscript is a set-overlap measure: the predicted and reference sets both contain exactly $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$ members, so precision, recall and F_1 take the same value by construction. That quantity is well defined and we report it as overlap, but it cannot distinguish a predictor that finds the critical set from one that merely produces a set of the right size. Table S16 reports three quantities that can, at operating points where precision and recall are free to differ.

The LOSO block reproduces the paper’s central negative finding on a different family of metrics: **HGT-QoS** ($F_1@7 = 0.383$, PR-AUC 0.471), **GAT-S-w** (0.384, 0.460) and **Topo-QoS** (0.358, 0.469) are indistinguishable, and **HGT** attains the best PR-AUC of the seven (0.477). Identification therefore separates the predictors no better than ranking does. The real-world block is where the two harnesses disagree: there the learned model leads on both measures by a wide margin (0.473 and 0.713 against 0.29–0.33 and 0.47–0.52), consistent with the full-population transfer result of Table 9 of the main manuscript and subject to the same caveat, that five systems cannot establish it.

Table S16: Identification quality at operating points where precision and recall are free to differ, so that none of these columns is the top- K overlap measure reported as $\text{Overlap}@K$ elsewhere. $F_1@7$ scores the top- K prediction against the labels’ own critical set ($I^*(v) \geq 0.5 \max I^*$); PR-AUC is threshold-free; nDCG@10 is rank-weighted. LOSO figures are means over the twelve inductive folds; real-world figures are means over the five transcribed systems, scored zero-shot against the same oracle.

Predictor	$F_1@7$	PR-AUC	nDCG@10
<i>Inductive LOSO, twelve folds</i>			
TOPO	0.336	0.442	0.522
TOPO-QoS	0.358	0.469	0.524
RM	0.302	0.403	0.469
GAT-S	0.294	0.388	0.409
GAT-S-w	0.384	0.460	0.498
HGT	0.372	0.477	0.553
HGT-QoS	0.383	0.471	0.501
<i>Zero-shot transfer, five open-source systems</i>			
RM	0.287	0.521	—
Topo	0.329	0.474	—
Topo-QoS	0.329	0.474	—
HGT-QoS	0.473	0.713	—

Rendered by `reproduce/render_table.py` from
`results/loso_all_variants_v5.json` and `results/realworld_zeroshot_v7.json`.

S19. The 2×2 on a Variance-Stabilized Scale

Scope note. This section analyzes the *unmatched* 2×2 of Table S27 (Section S31). The capacity- and channel-matched control (Table S26 of the main manuscript) removes the interaction and the typing effect entirely, so these robustness checks establish only that the unmatched interaction is not a metric or seed artifact, not that it reflects typing.

Spearman ρ is bounded on $[-1, 1]$ and compresses toward either end. A difference of differences computed on it is therefore not scale-free: two mechanisms that each move a predictor toward the attainable ceiling can appear sub-additive even if they contribute independently under any monotone rescaling. Since the sub-additivity claim of Section S31 *is* a claim about an interaction, it has to survive the transformation that removes that artifact.

Table S17 repeats the three orthogonal quantities of the 2×2 with each fold’s ρ passed through arctanh before the contrasts are formed. No cell of the shipped artifact reaches $|\rho| = 1$, so the transform is finite throughout. The interaction does not weaken; it grows, from -0.199 to -0.233 , remains negative on all twelve folds, and its bootstrap interval continues to exclude zero. Both main effects likewise survive. Sub-additivity on this corpus is therefore a property of the two mechanisms and not of the metric, which is what licenses reading typing and the QoS edge channel as substitutes — subject, as in the main text, to the capacity and directionality confounds that the unrun control arms would separate.

Table S17: The three orthogonal quantities of the 2×2 on the raw Spearman scale and on the Fisher z scale, Holm-corrected within each block of three. Main effects average over the other factor’s levels. **Won** counts folds with $\Delta > 0$; the interaction is negative on all twelve under both transformations. Post-hoc and exploratory, as in the main text.

Quantity	Δ (raw ρ)	Δ (Fisher z)	Won	p_{Holm}
Typing (main effect)	+0.134	+0.186	12/12	0.0015
QoS channel (main effect)	+0.187	+0.269	11/12	0.0015
Typing $\times$ QoS interaction	−0.199	−0.233	0/12	0.0015
<i>Interaction bootstrap 95% CI: [−0.258, −0.147] raw; [−0.309, −0.165] on z</i>				

S20. In-Distribution Per-Scenario Results

This table was moved out of the body because no comparison can be drawn down its columns: the typed and homogeneous arms read different substrates in-distribution (Section 5.2 of the main manuscript), so a difference between them confounds message passing with multi-entity visibility. It is retained because the per-scenario cells expose within-predictor fitting behavior that the fold means hide.

Table S18: In-distribution held-out evaluation across all twelve distributed architecture scenarios, reporting both Spearman rank correlation (ρ) and critical-set identification (Overlap@ K , with $K = \text{round}(0.20 \cdot n)$): mean over five seeds; n = held-out Application test count. Overlap@ K is top- K set overlap throughout (Section 5.3 of the main manuscript). Substrates differ in-distribution and are listed per predictor in Table S32. Each seed redraws the 60/20/20 split and model initialization. The held-out set is small on several scenarios — $n = 5$ on ATM, giving $K = 1$ — so a single rank swap moves ρ substantially and the per-scenario cells on the smaller topologies are coarsely quantized; they are reported for completeness and the row means, not individual cells, carry the comparison.

Scenario	n	Topo		Topo-QoS		GAT-S-P		GAT-S-P-w		HGT		HGT-QoS	
		ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1
ATM System	5	0.538	0.400	0.557	0.400	−0.393	0.000	−0.080	0.000	0.492	0.600	0.348	0.400
AV System	16	0.188	0.333	0.797	0.533	0.816	0.400	0.465	0.267	0.637	0.533	0.558	0.533
Enterprise	60	0.443	0.600	0.793	0.600	0.779	0.583	0.481	0.433	0.861	0.600	0.878	0.600
Financial Trading	12	0.387	0.200	0.512	0.400	0.565	0.400	0.666	0.400	0.693	0.500	0.730	0.500
Healthcare	10	0.291	0.200	0.399	0.000	0.725	0.300	0.575	0.300	0.575	0.400	0.607	0.500
Enterprise Integration (ESB)	14	0.179	0.267	0.429	0.400	0.363	0.467	−0.156	0.067	0.421	0.400	0.476	0.400
Industrial SCADA	28	0.601	0.533	0.710	0.533	0.656	0.533	0.478	0.500	0.787	0.667	0.839	0.633
IoT Smart City	40	0.320	0.350	0.397	0.350	0.580	0.450	0.538	0.425	0.849	0.650	0.850	0.650
Logistics Fleet	22	0.511	0.500	0.652	0.400	0.746	0.500	0.780	0.550	0.796	0.550	0.815	0.500
Microservices (synthetic)	18	0.219	0.150	0.344	0.250	0.351	0.400	0.363	0.450	0.141	0.300	0.664	0.600
Real-Time Gaming	15	0.360	0.533	0.802	0.533	0.464	0.333	0.471	0.400	0.651	0.533	0.641	0.400
Telecom RAN	24	0.402	0.480	0.422	0.280	0.608	0.320	0.350	0.360	0.591	0.280	0.526	0.320
Mean	—	0.370	0.379	0.568	0.390	0.522	0.391	0.411	0.346	0.624	0.501	0.661	0.503

S21. Results at a Glance

The results-at-a-glance figure now appears in the main manuscript as Figure 4, generated from the same artifacts as Tables 6 and S26 of the main manuscript (`reproduce/render_headline_figure.py`). It omits the typing $\times$ QoS interaction of the unmatched 2×2 (Table S27), which the capacity-matched control shows to be an effect of capacity and channel width (Section S31); oracle agreement is reported in Section S12.

S22. Illustrative Diagnostic Remediation Card

The card below is composed by hand from the ATM topology to show the shape of the explanation layer’s output. It is an illustration of the format, not a measured export, and no claim rests on it.

Table S19: Illustrative Diagnostic Remediation Card for a flagged component (**TransactionProcessor** in the ATM System case study). The explanation layer maps structural metrics to ISO/IEC sub-characteristics to guide engineering intervention.

Diagnostic Attribute	Evaluated Value	Structural Interpretation
Component ID	TransactionProcessor	Core transaction coordinator (V_{app})
Tukey Risk Tier	CRITICAL	$Q(v) = 0.84 > Q_3 + 1.5 \cdot \text{IQR}$
<i>ISO/IEC 25010 Structural Quality Profile Breakdown</i>		
Availability (A)	0.89 [High SPOF Risk]	$\text{AP}_c^{\text{dir}} = 1.00$, $\text{QSPOF} = 0.92$, $\text{CDI} = 0.78$
Fault Tolerance (FT)	0.31 [Low Cascade Propagation]	$\text{RPR} = 0.25$, $\text{Deg}_{\text{in}} = 0.30$, $\text{CDPot} = 0.38$
Maintainability (M)	0.65 [Moderate Structural Coupling]	$\text{BT} = 0.70$, $w_{\text{out}} = 0.60$, $\text{CQP} = 0.40$
Structural Diagnosis	Isolated Single Point of Failure (Severe A deficit, low downstream cascade reach)	
Prescriptive Action	Deploy warm-standby replica and establish redundant message broker routing	

S23. General Multi-Task Objective

The implementation supports a more general objective than the one optimized in every reported run (Eq. 3 of the main manuscript):

$$\mathcal{L} = \mathcal{L}_{\text{composite}} + 0.5 \cdot \mathcal{L}_{\text{dimension}} + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} + \lambda_{\text{RM}} \cdot \mathcal{L}_{\text{consistency}}, \quad (6)$$

where $\mathcal{L}_{\text{dimension}} = \sum_d \frac{1}{m_d} \sum_{d \in \{R, M\}} m_d \cdot \text{MSE}(\hat{d}(v), d^*(v))$ uses a boolean dimension mask $m = [m_R, m_M]$, and $\mathcal{L}_{\text{consistency}}$ is an optional mean-squared alignment of the two auxiliary heads with the explanation layer’s R and M scores on unlabeled nodes. Every reported run sets $\lambda_{\text{RM}} = 0$, so the predictor and the explanation layer share no parameters, and $m = [1, 0]$ with $R^*(v) \equiv I^*(v)$, because no simulated maintainability label exists. Under these settings $\mathcal{L}$ reduces exactly to the active objective. The code also provides a domain reweighting $\hat{Q}_{\text{domain}}(v) = q_R \hat{a}_1(v) + q_M M_{\text{static}}(v)$ for an ISO/IEC 25019 context-of-use vector; it is not evaluated in this study.

S24. Seed-Aggregation Robustness of the 2×2

Scope note. This section analyzes the *unmatched* 2×2 of Table S27 (Section S31). The capacity- and channel-matched control (Table S26 of the main manuscript) removes the interaction and the typing effect entirely, so these robustness checks establish only that the unmatched interaction is not a metric or seed artifact, not that it reflects typing.

GAT-S, the floor cell of the reported 2×2 , has a median within-fold seed standard deviation of 0.298, against 0.114 (HGT), 0.024 (GAT-S-w) and 0.052 (HGT-QoS). If a few collapsed seeds pull its fold means down, both main effects and the interaction inherit that optimization failure. Table S20 re-forms each fold score from the per-seed results of the same twelve-fold artifact under three aggregations and recomputes the three orthogonal quantities, Holm-corrected across the three within each aggregation. No model is retrained (`reproduce/factorial_seed_robustness.py`; `results/factorial_seed_robustness_v5.json`).

The interaction shrinks by a sixth (median) to a quarter (trimmed) and stays negative on every fold, and the QoS-channel main effect shrinks most. Collapsed GAT-S seeds therefore inflate the reported effects but do not produce them. The check cannot address under-training that affects every seed alike, which requires per-arm tuning under an equal budget.

S25. Closed-Form Scores With the Articulation Term Restored

The closed-form scores are specified as $0.6 \cdot \text{BT} + 0.4 \cdot \text{AP}$ (Section 5.2 of the main manuscript). In the evaluated implementation the articulation term reads zero for every node, because the cached structural metrics carry no articulation score. Table S21 recomputes both scores on the same twelve holdouts, Application

Table S20: The 2×2 quantities of Table S27 under three seed aggregations. *Mean* reproduces the reported table; *median* is robust to one or two collapsed seeds per fold; *trimmed* drops each arm’s worst seed in every fold before averaging. CI: bootstrap over folds ($B = 2,000$).

Aggregation	Quantity	$\Delta\rho$	Won	p	p_{Holm}	95% CI
Mean	Typing (main)	+0.134	12/12	0.0005	0.0015	[+0.095, +0.176]
	QoS channel (main)	+0.187	11/12	0.0015	0.0015	[+0.108, +0.244]
	Interaction	−0.199	0/12	0.0005	0.0015	[−0.258, −0.147]
Median	Typing (main)	+0.118	11/12	0.0010	0.0020	[+0.067, +0.175]
	QoS channel (main)	+0.145	11/12	0.0093	0.0093	[+0.070, +0.206]
	Interaction	−0.168	0/12	0.0005	0.0015	[−0.231, −0.112]
Trimmed	Typing (main)	+0.118	11/12	0.0010	0.0020	[+0.077, +0.160]
	QoS channel (main)	+0.128	11/12	0.0093	0.0093	[+0.060, +0.179]
	Interaction	−0.146	0/12	0.0005	0.0015	[−0.177, −0.115]

population and $I^*(v)$ labels, with the articulation term taken from the articulation severity SaG computes on the projection (`reproduce/topo_ap_sensitivity.py`; `results/topo_ap_sensitivity.json`). The *as run* columns reproduce the published per-fold values exactly (maximum absolute difference $< 10^{-15}$).

Table S21: Spearman ρ of the closed-form scores per LOSO holdout, as run (articulation term zero) and with the articulation term restored. n_{AP} counts nodes with non-zero articulation severity on the projection.

Holdout	n_{AP}	Topo as run	Topo AP restored	Topo-QoS as run	Topo-QoS AP restored
ATM	1	0.256	0.083	0.311	0.132
AV System	0	0.468	0.468	0.753	0.753
Enterprise	0	0.431	0.431	0.795	0.795
Financial Trading	0	0.237	0.237	0.586	0.586
Healthcare	0	0.077	0.077	0.369	0.369
Enterprise Integration (ESB)	0	0.112	0.112	0.430	0.430
Industrial SCADA	10	0.608	0.563	0.650	0.609
IoT Smart City	6	0.251	0.234	0.351	0.331
Logistics Fleet	2	0.576	0.578	0.741	0.742
Microservices	0	0.229	0.229	0.265	0.265
Real-Time Gaming	0	0.377	0.377	0.810	0.810
Telecom RAN	0	0.562	0.562	0.576	0.576
Mean		0.349	0.329	0.553	0.533

Articulation points are rare on these projections (four of twelve scenarios have any), and where they exist the binary term lowers the ranking correlation. The evaluated, betweenness-only form is therefore the stronger reference, and every comparison in the main manuscript is made against it.

S26. Hybrid-HGT Per-Fold Results

Table S22 lists the per-fold Spearman ρ behind Table 6 of the main manuscript: CPU sweeps, twelve LOSO folds, fold score = mean over five seeds, Application population (`results/loso_hybrid_cpu.json`, `results/loso_hybrid_gat_cpu.json`). The Topo-QoS column is bit-identical across the two sweeps. Folds are ordered by the closed-form engine’s own score. Each hybrid outperforms Topo-QoS on every fold except Enterprise, and keeps most of the closed-form engine’s strength on the folds where its underlying learned engine loses to it.

Table S22: Per-fold LOSO ρ for SaG’s closed-form engine, the two learned engines and their hybrids (CPU sweeps; bold marks the best in each row).

Holdout	Topo-QoS	HGT-QoS	Hybrid-HGT	GAT-QoS	Hybrid-GAT
Real-Time Gaming	0.810	0.789	0.837	0.685	0.825
Enterprise	0.795	0.426	0.735	0.407	0.768
AV System	0.753	0.704	0.782	0.732	0.793
Logistics Fleet	0.741	0.771	0.792	0.654	0.806
Industrial SCADA	0.650	0.684	0.758	0.721	0.768
Financial Trading	0.586	0.695	0.754	0.713	0.797
Telecom RAN	0.576	0.427	0.648	0.574	0.656
Enterprise Integration (ESB)	0.430	0.548	0.564	0.630	0.568
Healthcare	0.369	0.730	0.625	0.798	0.686
IoT Smart City	0.351	0.688	0.590	0.720	0.654
ATM	0.311	0.523	0.429	0.506	0.447
Microservices	0.265	0.475	0.366	0.479	0.429
Mean	0.553	0.622	0.657	0.635	0.683

S27. A Proposed Explanation Layer (Not Evaluated)

A ranking says where risk is highest, not how to reduce it. A component may be critical because it is an unreplicated single point of failure, an error-propagating cascade hub, or a high-coupling maintainability bottleneck, and each calls for a different intervention: replication, circuit breakers, or decoupling. SaG includes a layer that attributes a flagged component to one of these causes. **This layer is a design proposal. Its attributions have not been validated against practitioner judgment or against the outcome of applying the remediation it names, and nothing in the main manuscript evaluates it.** It is described here, rather than in the main manuscript, because it is part of the released tool and its inputs overlap the rankers’ features, but it is not one of the paper’s contributions. It is not used as a ranker; as a ranker it is weak (its composite reaches $\rho = 0.234$ against I^* at the shipped shrinkage $\lambda = 0.70$, 0.200 with the raw AHP weights and 0.319 with uniform intra-dimension weights; Section S4).

Following ISO/IEC 25010:2023 and ISO/IEC 25019:2023, the layer profiles each component along **Fault Tolerance** (FT), which informs circuit breakers and redundancy; **Availability** (A), which informs replication; and **Maintainability** (M), which informs decoupling and refactoring (Figure S3). Three of the weight vectors below encode a declared priority vector rather than independent elicitation (Section S7), and moving the intra-dimension weights toward the elicited judgment lowers the layer’s rank correlation monotonically. Validating the attributions is the layer’s principal open question.

All metrics are rank-normalized to $[0, 1]$ within the graph and combined with AHP-derived weights (Section S7):

- $FT(v) = 0.45 \cdot \text{RPR}(v) + 0.30 \cdot \text{Deg}_{\text{in}}(v) + 0.25 \cdot \text{CDPot}_{\text{enh}}(v)$, over Reverse PageRank, normalized in-degree and normalized cascade depth on G_{analysis}^T ;
- $A(v) = 0.25 \cdot \text{AP}_c^{\text{dir}}(v) + 0.20 \cdot \text{QSPOF}(v) + 0.20 \cdot \text{BR}(v) + 0.25 \cdot \text{CDI}(v) + 0.10 \cdot w(v)$, over directed articulation severity, QoS-weighted SPOF severity, bridge ratio, the Connectivity Degradation Index and the node QoS weight;
- $R(v) = r_{\text{FT}} \cdot FT(v) + (1 - r_{\text{FT}}) \cdot A(v)$ with $r_{\text{FT}} = 0.36$;
- $M(v) = 0.35 \cdot \text{BT}(v) + 0.30 \cdot w_{\text{out}}(v) + 0.15 \cdot \text{CQP}(v) + 0.12 \cdot \text{CouplingRisk}_{\text{enh}}(v) + 0.08 \cdot (1 - \text{CC}(v))$, over betweenness, QoS-weighted efferent coupling, code-quality penalty, coupling risk and clustering.

The composite is $Q(v) = 0.80 \cdot R(v) + 0.20 \cdot M(v)$, and an ISO/IEC 25019 context-of-use vector can reweight R and M . Intra-dimension weights are shrunk towards a uniform prior ($\lambda = 0.70$); a fully uniform prior ranks better against I^* (0.319, against 0.234 at $\lambda = 0.70$ and 0.200 for the raw AHP weights; Section S4).

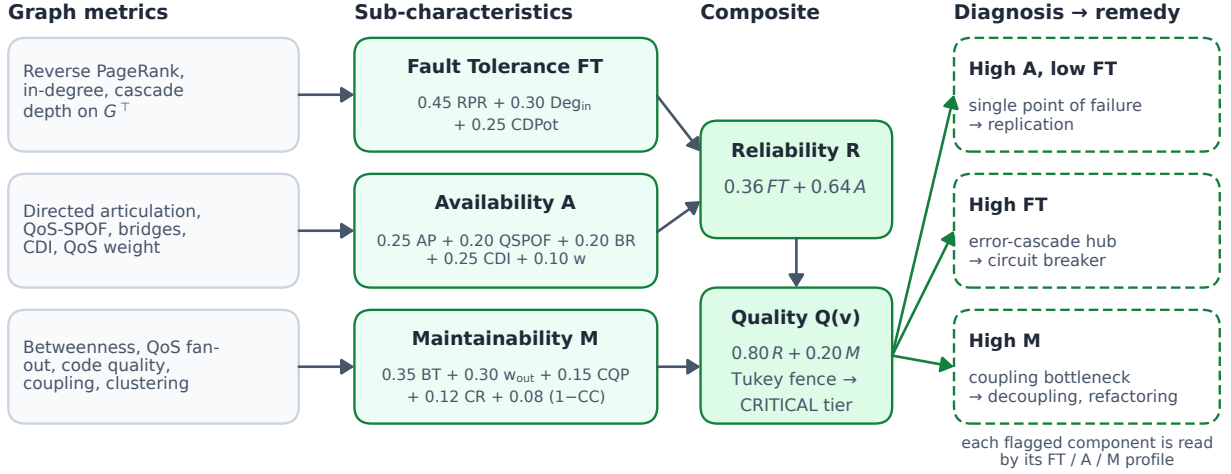


Figure S3: The proposed explanation layer (not evaluated). Rank-normalized graph metrics feed the ISO/IEC 25010 sub-characteristics Fault Tolerance, Availability and Maintainability (CR: coupling risk; CC: clustering coefficient), which combine into Reliability and the composite $Q(v)$. A component above the Tukey fence of Q is flagged, and its $FT/A/M$ profile names the remediation class.

Components above the Tukey upper fence of Q are flagged CRITICAL (mean 4.2% of components). High A with low FT is read as a single point of failure needing replication; high FT as a cascade hub needing circuit breakers (example card: Section S22). Generating candidate repairs and verifying them counterfactually is companion work and is not evaluated here.

S28. Analysis-Plan Amendment Log

The analysis plan (Section 5.3 of the main manuscript) was amended sixteen times, and one deviation from it is recorded. Table S23 lists the plan and every amendment, with its date, whether it was written before or after the results it governs, what it changed, the contrasts it registered, and its outcome. The full text, with each design and decision rule as committed, is `docs/research/jss/PREREGISTRATION.md` in the replication package.

Completeness. No amendment was issued and left out of this table. The log is under version control, and its history shows the plan and Amendments 1–15 added in order, one commit each, at commits 44713326 (plan and A1), fef9f211 (A2), f1162e90 (A3), c75fe944 (A4), e80c3bec (A5), a80008e2 (A6), 4878cf28 (A7), 0b3f7ca0 (A9), 0c52329d (A10), e981cf01 (A11), 8a30eab2 (A12), 18f44f44 (A13) and 714e70fc (A14 and the deviation record) and 2c5275f2 (A15). These timestamps are the repository’s own; the plan was not deposited with an independent registry; Amendment 12’s deviation notes were appended before the analyses they govern were run. Amendment 8 was first committed as Amendment 7 on a parallel branch and renumbered, with its text otherwise unchanged, when the branches were merged. No amendment was removed, and no registered design or decision rule was edited after its commit. Only two passages were edited later: the plan’s motivation paragraph, rewritten once the pre-revision artifact it cited was withdrawn, and the plan’s outcome note, which is marked as recorded after the fact.

Registered arms. All four model arms that Amendment 2 registered have now run. GAT and GAT-QoS form, with HGT and HGT-QoS, the matched 2×2 of the main manuscript. The two later arms were each run in their own CPU sweep, with HGT-QoS re-run in the same invocation. GAT-w, a capacity-matched untyped GAT that reads only the scalar QoS weight $w(e)$, reaches $\rho = 0.633$ (registered contrast HGT-QoS vs. GAT-w -0.011 , HGT-QoS ahead on 5/12 folds, $p = 0.68$). HGT-QoS-U, which drops HGT’s reverse pass and with it the only route by which any message reaches an Application, reaches 0.632 (HGT-QoS vs. HGT-QoS-U -0.010 , 6/12, $p = 0.91$), and transfers zero-shot at 0.804 against 0.760, higher on all five system models. Neither

control changes a decision: on the raw multigraph, typed parameters, the 16-D edge width and message passing into Applications each add nothing measurable (main Section 7.2). On the dependency graph, where messages come from each component’s dependents, message passing does add (Section S39). Amendment 2’s label-side arm, a LOSO sweep against a label cache with the oracle’s QoS ladder disabled, was not run as a sweep. Section 4.3 of the main manuscript bounds the label’s QoS content instead, using the rank agreement between labels computed with and without the ladder.

Predictor names. The plan and its amendments name predictors with the earlier labels; this log uses the main manuscript’s names, and Table S33 maps the two.

Research-question numbering. The plan and its amendments number five research questions. The main manuscript consolidates them into four. The plan’s RQ1 and RQ2 keep their numbers. Its RQ3 (QoS encoding and robustness) is answered within the main manuscript’s RQ2 and threats to validity. Its RQ4 (transfer) and RQ5 (cost) are the main manuscript’s RQ3 and RQ4. RQ numbers in Table S23 follow the plan.

Omnibus correction. Each registration Holm-corrects only its own contrasts. The sequence is adaptive: Amendment 6 was proposed because of Amendment 2’s result. Table S24 therefore pools all thirteen registered contrasts under one Holm correction (`reproduce/omnibus_holm.py`, `results/omnibus_registered_holm.json`). Both hybrid primaries stay below $\alpha = 0.05$, and no other registered contrast is close. As everywhere in this study, the p -values are nominal, because LOSO folds share training scenarios.

S29. LOSO Active-Stratum Results

Between 21% (Microservices) and 52% (Healthcare) of each held-out Application population carries exactly zero simulated impact, so a predictor can score well by separating components that can propagate a failure from those that cannot, without ordering the propagating ones correctly. Table S25 re-scores all twelve LOSO folds of Table S34 on the *active stratum* — the $n_{>0}$ components with strictly positive impact — using the same predictions, folds and seeds.

The restriction costs every predictor roughly half its correlation, learned or not: retained fractions run from 49% (RM) to 56% (HGT-QoS) with no systematic separation between training-free and learned families, so roughly half of every predictor’s full-population correlation reflects separating inert components from active ones. The method ordering is unchanged. No significance test is reported on this stratum, so it supports the full-population ordering descriptively and adds no new verdict.

S30. The Capacity-Matched 2×2 on the Raw Multigraph

Table S26 is the capacity- and channel-matched 2×2 over relation typing and the “QoS” inputs that Section 6.2 of the main manuscript summarizes. It runs on the raw multigraph, where forward messages never reach Applications, so it compares per-node models and carries no information about relational message passing; Table S70 (F4) repeats it with w_{in} held in both arms.

S31. The Unmatched (Naive) 2×2

Table S27 is the 2×2 over relation typing and the QoS edge channel formed by the four learned arms of Table S34. The untyped arms there have 28,168 parameters against 434,620 for HGT and read a 1-dimensional edge channel against HGT-QoS’s 16 dimensions. The capacity- and channel-matched control (Table S26 of the main manuscript) removes both differences, and at matched capacity the same untyped design reaches 0.563 rather than 0.317. The +0.234 typing gain and -0.199 interaction below are therefore effects of capacity and channel width, and this table is not used for any claim about typing. Neither the Fisher- z transform nor robust seed aggregation (Sections S14 and S21), which left the unmatched interaction intact, could detect this confound: both hold the four unmatched arms fixed.

Table S23: The analysis plan, its sixteen amendments and one recorded deviation. Status tier (main §5.3): the plan is confirmatory; amendments written before any result of their own arms are registered secondary; the rest are exploratory. “Before” means no outcome of the analysis the entry governs existed when it was committed. Contrast counts are the decision-bearing contrasts the entry registered; the exploratory analyses it added are not counted.

Entry	Date	Written	What it fixed or changed	Registered contrasts	Outcome (where reported)
Plan	2026-09-06	Before any twelve-fold result	Primary HGT-QoS vs Topo-QoS; secondary HGT vs Topo-QoS; LOSO, five seeds, fold as unit; nested selection rule	2, Holm within	Neither significant (+0.085, Holm $p = 0.303$) (main §6.1)
A1	2026-09-06	Before any outer result	Inner selection on a two-fold inner holdout instead of full inner LOSO (compute budget)	None	Not applied: the published learned results use one fixed configuration (see the deviation row)
A2	2026-09-12	Before any control result	Capacity-, edge-channel- and directionality-matched control arms, a label-side arm, and a four-row decision rule for the typing claim	7 (matched 2×2 and four controls), Holm within	“Margin does not survive”: typing claims withdrawn; QoS channel +0.073 (main §6.2)
A3	2026-09-13	Before any v5 result	Re-baseline as the v5 sweep after artifact drift; withdrawal of an unbacked control-arm paragraph; exploratory GBM baseline, architecture contrasts, contrast CIs, extra populations	None	v5 is the reported sweep; the withdrawn paragraph is not reported (main §6.1–6.2)
A4	2026-09-20	After the v5 results	Peer-review revision: “registered” replaces “pre-registered”; RQ4 re-run at the common 3-layer, 300-epoch protocol; Fisher- z 2×2 , real-system CIs, cost-ratio table, drift ledger, energy bound, all exploratory	None	Reported as exploratory (main §5.3, §6.3, §6.4, §7.3–7.4)
A5	2026-09-23	Before any hybrid result	Hybrid-HGT (Topo-QoS prior on HGT-QoS); CPU sweep; decision rule	2, Holm within	Primary significant (+0.103, Holm $p = 0.0068$): headline hybrid (main §6.1, Section S26)
A6	2026-09-24	Before any result, but after A2’s result	Hybrid-GAT (same prior on GAT-QoS); decision rule including a transfer criterion	2, Holm within	Primary significant (+0.130, Holm $p = 0.0029$); transfers less well ($0.662 < 0.695$), so it does not replace Hybrid-HGT (main §6.1)
A7	2026-09-25	Before any result	Training-free dependency counts (InDeg, Reach, Reach-QoS, CDI) and three QoS-attribution controls (constant topic weight, permuted QoS, QoS-independent corpus); oracle-parameter sweep; inert-vs-active rule; decision rules R1, R2, R2’, R3	4, Holm within	R1, R2 and R2’ all apply: InDeg 0.764 exceeds every raw-multigraph learned engine; the closed-form gain is attributed to QoS-weighted dependency multiplicity, not to QoS content (main §6.1; Section S38)
A8	2026-09-26	After its results	Attribution controls: disclosure that the untyped GATs score Applications per node on the raw multigraph; GAT-QoS-nf, GBM-Feat-QoS, and the first run of A3’s GBM-Feat; receptive-field probe	None (5 exploratory, Holm within)	QoS gain carried by node features (+0.095), not the edge channel (−0.023) (main §6.2)
A9	2026-09-26	Before any result of its learned arms	The same learners on the dependency graph (GAT-P, GAT-P-QoS, GAT-P+InDeg with an InDeg prior, HGT-P-QoS); decision rules D1–D3, M, Z	12, Holm within	D3, M and Z apply: no arm exceeds InDeg, the two QoS arms match it, and the GAT arms gain +0.075 to +0.113 over their raw-multigraph counterparts (main §6.2; Section S39)
A10	2026-09-26	Before any result	The value of the dependency derivation: InDeg against raw-graph counts, Reach against Rule-1-only reach; two identities recorded; decision rules E1–E3	3, Holm within	E1 and E2 apply. As corrected in round 7 (Pubs-raw had been scored on publishers only): InDeg beats raw degree by +0.565 and topics published by +0.033 (12/12 and 11/12); Rule 5 adds +0.058 to reach (main §6.1; Section S39)
A11	2026-09-26	Before any result	Full-population, five-seed I_{dyn} labels; a learned combination of training-free signals (GBM, four arms); cross-oracle robustness families A–C	6, Holm within three families	Completed after round 7 (an earlier attempt was killed by the out-of-memory killer). A’’: no robustness gain over the first-order expansion (−0.121, Holm $p = 0.001$); B: the learned I_{dyn} approximation beats it on I_{dyn} (+0.093, Holm $p = 0.019$); C: QoS features help on I_{dyn} only (+0.095, Holm $p = 0.014$; carried by the declared-rate columns, A15; the oracle reads no payload); R: full-population I_{dyn} replaces the $n = 30$

Table S24: Every registered contrast under its own family’s Holm correction and under one Holm correction over all thirteen. Two-sided Wilcoxon signed-rank over twelve LOSO folds. The plan’s contrasts come from the v5 sweep (Table S34), and the amendments’ contrasts from the CPU sweeps that registered them. Bold marks omnibus-adjusted $p < 0.05$.

Entry	Contrast	$\Delta\rho$	p	p_{Holm} (family)	p_{Holm} (omnibus)
A6	Hybrid-GAT vs Topo-QoS	+0.130	0.0015	0.0029	0.0190
A5	Hybrid-HGT vs Topo-QoS	+0.103	0.0034	0.0068	0.0410
A2	QoS edge channel (main effect)	+0.073	0.0425	0.1274	0.4673
Plan	HGT-QoS vs Topo-QoS	+0.085	0.1514	0.3027	1.000
A6	Hybrid-GAT vs GAT-QoS	+0.048	0.3013	0.3013	1.000
A2	HGT-QoS vs GAT-QoS	−0.013	0.3804	1.000	1.000
Plan	HGT vs Topo-QoS	−0.002	0.4697	0.4697	1.000
A2	Typing (main effect)	−0.014	0.4697	0.9395	1.000
A2	HGT vs GAT	−0.015	0.5693	1.000	1.000
A2	HGT-QoS vs GAT-w	−0.011	0.6772	1.000	1.000
A2	Typing $\times$ QoS interaction	+0.001	0.7334	0.9395	1.000
A5	Hybrid-HGT vs HGT-QoS	+0.035	0.7334	0.7334	1.000
A2	HGT-QoS vs HGT-QoS-U	−0.010	0.9097	1.000	1.000

Table S25: LOSO means over twelve folds on the full Application population (ρ) and restricted to components with strictly positive simulated impact ($\rho_{>0}$). Same predictions, folds, and seeds; only the evaluated subset differs. Retained is defined as the active-stratum ratio $\rho_{>0}/\rho$ (percentage of full-population correlation preserved).

Predictor	ρ (full)	$\rho_{>0}$ (active)	Retained
RM / $Q(v)$	0.205	0.102	49%
Topo	0.349	0.181	52%
Topo-QoS	0.553	0.280	51%
GAT-S	0.317	0.159	50%
GAT-S-w	0.604	0.328	54%
HGT	0.551	0.299	54%
HGT-QoS	0.638	0.356	56%

Table S26: The 2×2 factorial design with capacity and edge-channel width matched: GAT ($\neg T \neg Q$, 437,496 parameters), HGT ($T \neg Q$, 434,620), GAT-QoS ($\neg T Q$, 429,992), HGT-QoS ($T Q$, 434,620). One CPU sweep, twelve LOSO folds, five seeds, Application population. Holm correction across the three orthogonal quantities. Cell means: 0.563, 0.548, 0.635, 0.622.

Quantity	Contrast	$\Delta\rho$	95% CI	Won	W	p	p_{Holm}
<i>Three orthogonal quantities, Holm-corrected across these three</i>							
Typing (main effect)	averaged over Q	−0.014	[−0.052, +0.023]	4/12	29.0	0.470	0.940
“QoS” inputs (main effect)	averaged over T	+ 0.073	[+0.013, +0.120]	10/12	13.0	0.043	0.127
Typing $\times$ “QoS” interaction	difference of differences	+0.001	[−0.050, +0.042]	6/12	34.0	0.733	0.940
<i>Simple effects — descriptive, not separately corrected</i>							
Typing, QoS absent	HGT vs. GAT	−0.015	[−0.064, +0.033]	5/12	31.0	0.569	—
Typing, QoS present	HGT-QoS vs. GAT-QoS	−0.013	[−0.054, +0.026]	4/12	27.0	0.380	—
“QoS” inputs, typing absent	GAT-QoS vs. GAT	+ 0.072	[+0.028, +0.109]	10/12	9.0	0.016	—
“QoS” inputs, typing present	HGT-QoS vs. HGT	+0.073	[−0.002, +0.136]	10/12	19.0	0.129	—

Table S27: *Naïve* 2×2 over relation typing (T) and the QoS edge channel (Q), unmatched in capacity ($15.4\times$) and edge-channel width; superseded for all typing claims by Table S26 of the main manuscript. Cells are the four reported learned arms: GAT-S ($\neg T \neg Q$), HGT ($T \neg Q$), GAT-S-w ($\neg T Q$), HGT-QoS ($T Q$). **Holm correction is applied across the three orthogonal quantities in the upper block only**; the four simple effects below are algebraically determined by those three and are reported descriptively. Main effects average over the other factor’s levels; the typing effect reflects the joint transition to HGT and is confounded with capacity and channel width. **Won** counts folds with $\Delta > 0$; the interaction is *negative* on all twelve. All quantities are post-hoc, and none were registered in advance. Fold overlap makes the reported p -values nominal (Section 7.4 of the main manuscript).

Quantity	Contrast	$\Delta\rho$	Won	W	p	p_{Holm}
<i>The 2×2: three orthogonal quantities, Holm-corrected across these three</i>						
Typing (main effect)	averaged over Q	+0.134	12/12	0.0	0.0005	0.0015
QoS channel (main effect)	averaged over T	+0.187	11/12	2.0	0.0015	0.0015
Typing $\times$ QoS interaction	difference of differences	−0.199	0/12	0.0	0.0005	0.0015
<i>Simple effects — descriptive, not separately corrected</i>						
Typing, QoS absent	HGT vs. GAT-S	+0.234	12/12	0.0	0.0005	—
Typing, QoS present	HGT-QoS vs. GAT-S-w	+0.035	9/12	19.0	0.1294	—
QoS channel, typing absent	GAT-S-w vs. GAT-S	+0.287	11/12	1.0	0.0010	—
QoS channel, typing present	HGT-QoS vs. HGT	+0.087	10/12	22.0	0.2036	—

S32. Zero-Shot Transfer: Bootstrap Intervals and Active Stratum

Table S28 complements Table 9 of the main manuscript with percentile bootstrap intervals over the five system models, on the full Application population and on its active stratum.

Table S28: Means over the five system models on the full Application population (ρ) and restricted to components with strictly positive simulated impact ($\rho_{>0}$), with percentile bootstrap intervals over the five systems ($B = 2,000$). All four predictors are scored on identical labels, populations and node sets from one run, so the columns are commensurable. At $n = 5$ the intervals are descriptive and carry no significance claim. **The full-population intervals separate the learned model from all three baselines; every active-stratum interval spans zero**, which is why RQ4 is reported as established on the full population and unresolved on the active one.

Predictor	ρ (full), 95% CI	$\rho_{>0}$ (active), 95% CI
RM / $Q(v)$	0.516 [0.343, 0.680]	−0.055 [−0.292, +0.213]
Topo	0.511 [0.346, 0.703]	−0.083 [−0.269, +0.108]
Topo-QoS	0.526 [0.357, 0.699]	−0.092 [−0.268, +0.094]
HGT-QoS	0.760 [0.714, 0.819]	+0.236 [−0.053, +0.525]

Restricted to components that propagate failures, HGT-QoS keeps a positive mean correlation where closed-form centrality turns negative, but its interval spans zero, as do the centrality baselines’, so this comparison is unresolved at five systems. Transitive reach on the dependency graph keeps $\rho_{>0} = 0.871$ on the same components (Table S42). $\rho_{>0}$ is positive on the three models of publish–subscribe systems and non-positive on the two modeled after RPC systems. Both RPC-derived models are encoded with topics and brokers and labeled by the same forward-reachability oracle, so this split cannot be attributed to call-tree semantics.

S33. Analysis Gate Against Direct Simulation

Timing the cascade reachability labeling sweep (five seeds, node types Application/Broker/Library, the full labeling run) over all twelve scenarios, on the same machine, at the same commit and in the same measurement session as the gate, gives 0.08–4.49s per scenario against 0.16–79.3s for the analysis gate. Both maxima occur in the 520-component Enterprise mesh, so the largest scenario compares 4.5s of simulation against 79.3s of static analysis: there the gate costs roughly eighteen times as much as the simulation. Pairing the sweeps scenario by scenario (Table S29) gives $2.0\text{--}17.7\times$, median $5.6\times$: the gate is more expensive

on all twelve scenarios, but the typical premium is far below the maximum. The ratio tracks the derived projection’s size rather than the component count ($\rho = 0.951$ with $|E_{\text{proj}}|$, against 0.792 for cost against $|V|$). These are wall-clock figures on one commodity CPU, read to one significant figure: across sessions the gate maximum ranges over 77–83s and the oracle maximum over 4.5–4.8s, giving $16.7\times$ and $17.7\times$ on two independently paired sessions.

Table S29: Static analysis gate against the cascade-reachability oracle, per scenario, from one paired measurement session. **Gate** is structural analysis plus the 18 anti-pattern detectors, whose own share is negligible (≤ 0.19 s everywhere). **Oracle** is the full five-seed labeling sweep over Application, Broker and Library nodes. $|E_{\text{proj}}|$ is the number of derived **DEPENDS_ON** edges in the Application–Library projection the analysis stage traverses, read from the committed cache. The ratio is $2.0\text{--}17.7\times$ with a median of $5.6\times$: the eighteen-fold figure quoted elsewhere in this section is the maximum, not the typical case. Ordering by $|E_{\text{proj}}|$ rather than by $|V|$ is what makes the column monotone — Enterprise carries 300 applications over only 120 topics, so its Rule-1 projection is near-complete, while IoT Smart City has more components and a sixth of the edges at a third of the cost.

Scenario	$ E_{\text{proj}} $	Gate (s)	Oracle (s)	Ratio
Enterprise	26,276	79.27	4.487	$17.7\times$
Enterprise Integration (ESB)	4,641	3.17	0.245	$12.9\times$
AV System	3,073	2.61	0.355	$7.4\times$
Financial Trading	2,657	1.71	0.239	$7.1\times$
Real-Time Gaming	2,318	2.43	0.379	$6.4\times$
Healthcare	1,590	0.92	0.149	$6.2\times$
IoT Smart City	1,835	5.48	1.104	$5.0\times$
Telecom RAN	2,116	3.70	1.065	$3.5\times$
Logistics Fleet	1,660	2.74	0.939	$2.9\times$
Microservices (synthetic)	1,524	2.29	0.828	$2.8\times$
Industrial SCADA	897	2.43	1.191	$2.0\times$
ATM System	155	0.16	0.080	$2.0\times$
Median	—	—	—	$5.6\times$

Breadth-first cascade traversal is cheaper than all-pairs connectivity degradation, so on raw CPU time direct simulation is faster wherever its parameters are available. SaG’s analysis additionally scores infrastructure components and dependency edges that node-level message-flow simulation leaves unscored, and its metrics are amenable to caching across commits (not implemented here).

Training cost. Training is a separate, one-off cost per model version: the last sweep with recorded per-arm wall-clock (CPU, sequential, 12 folds $\times$ 5 seeds = 60 fits per arm) took 0.6 h for **GAT-S**, 0.9 h for **GAT-S-w**, 1.4 h for **HGT** and 4.9 h for **HGT-QoS**, 7.7 CPU-hours in all. The GPU sweep behind Table S34 did not record per-fit durations.

S34. Latency and Scaling Micro-benchmarks

This section preserves the legacy profiling micro-benchmarks referenced in Section 6.4 of the main manuscript. Table S30 reports the per-stage latency breakdown across graph sizes from an earlier system-layer analysis session. Table S31 reports the scaling behavior of the training-free counting path against the published 5-seed labeling sweep of I^* up to 9,990 components. The definitive, like-for-like single-session benchmark across all stages is presented in Table 10 of the main manuscript; detailed execution scripts and raw profiling traces are archived in `docs/research/jss/experiments/rq4-cost.md`.

S35. Predictor Taxonomy and Substrate Details

Table S32 lists every predictor run in the study; the main manuscript’s taxonomy keeps only those it reports. A name gives the architecture and then what distinguishes it:

- **-QoS**: QoS-weighted distances for Topo, the 16-D QoS edge vector for a GNN;

Table S30: Per-stage latency of the inference pipeline across graph sizes (CPU, median of 3 runs; 5 for the forward pass; an earlier session, system-layer analysis). Only ratios within one row are meaningful: comparing this table’s analysis stage with Table S31’s oracle mixes stages, sessions and graphs. The analysis stage is stable across repeats (p10–p90 within 1% of the median), while the forward pass is dominated by interpreter and dispatch overhead; the 249-node row carries first-call warm-up.

		Analyze	Graph →	HGT	Analyze :	Forward
$ V $	$ E $	(s)	Tensor (s)	Forward (ms)	Forward	p10–p90 (ms)
249	1,127	1.74	0.010	26.5	66×	13.0–34.4
499	2,402	8.32	0.022	16.4	509×	15.6–16.4
999	6,422	44.54	0.056	21.1	2,108×	19.1–36.5
1,998	19,301	239.34	0.157	56.2	4,259×	43.8–57.8

Table S31: The counting path against the published five-seed labeling sweep of I^* over Application, Broker and Library nodes, on generated graphs of increasing size (Amendment 12, R6, an earlier session; CPU, median of 5 runs; I^* timed once at 4,995 components and not at 9,990, where one pass would take hours). Projection: deriving the Application–Library `DEPENDS_ON` graph from the manifest.

		Projection +	Reach	I^* sweep	I^* :
$ V $	$ E $	InDeg (ms)	(ms)	(s)	count
249	1,081	8.0	11.7	4.0	500×
499	2,437	11.2	24.9	15.1	1,351×
999	6,372	25.3	113.1	58.5	2,313×
1,998	19,242	109.7	490.8	346.6	3,159×
4,995	94,790	426.8	4,369.9	2,163.5	5,069×
9,990	348,277	1,095.3	23,302.4	not timed	—

- **-w**: a scalar QoS edge weight $w(e)$;
- **-S**: a small GAT (28k parameters);
- **-P**: the Application–Library derived dependency graph (Rules 1 and 5) instead of the raw multigraph;
- **Hybrid-X**: engine X corrected by the **Topo-QoS** prior (**GAT-P+InDeg** uses the **InDeg** reference prior).

Unsuffix GAT and GAT-QoS are matched to HGT’s parameter budget on the raw multigraph.

Table S32: Full taxonomy of every predictor run in this study, including the in-distribution flow-projection variants of Section S20.

Predictor	Evaluation Substrate	Typing	Edge Features	Parameters	Trained?	Empirical Role
Topo	G_{analysis} (Application layer only)	No	None	0	No	Standard unweighted centrality baseline
Topo-QoS	G_{analysis} (Derived <code>DEPENDS_ON</code>)	No	Scalar $w(e)$	0	No	SaG closed-form engine
Betweenness (proj.)	G_{analysis} (Derived <code>DEPENDS_ON</code>)	No	None	0	No	Unweighted reference (Amendment 7)
InDeg / Reach / Reach-QoS / CDI	G_{analysis} (Derived <code>DEPENDS_ON</code>)	No	None or $w(e)$	0	No	Dependency counts (Amendment 7); InDeg , Reach : references, not predictors (Amendment 13)
Degree-raw / Pubs-raw / Reach-R1	Raw multigraph / Rule 1 only	No	None	0	No	Derivation controls (Amendment 10); Pubs-raw , Reach-R1 : references (Amendment 13)
GAT-S-P	G_{analysis} (Derived <code>DEPENDS_ON</code>)	Homogeneous	None	28,168	Yes	In-distribution counterpart of GAT-S (Table S18)
GAT-S-P-w	G_{analysis} (Derived <code>DEPENDS_ON</code>)	Homogeneous	Scalar $w(e)$	28,168	Yes	In-distribution counterpart of GAT-S-w (Table S18)
GAT-S	Native Multigraph	Homogeneous	None	28,168	Yes	Untyped, unweighted GNN floor
GAT-S-w	Native Multigraph	Homogeneous	Scalar $w(e)$	28,168	Yes	Adds a scalar QoS weight
GAT	Native Multigraph	Homogeneous	None	437,496	Yes	Untyped, matched to HGT; no QoS channel
GAT-w	Native Multigraph	Homogeneous	Scalar $w(e)$	439,272	Yes	Capacity control: matched GAT with the scalar weight only
GAT-QoS	Native Multigraph	Homogeneous	16-D QoS Vector	429,992	Yes	Untyped, matched to HGT-QoS; SaG learned engine
GAT-QoS-nf	Native Multigraph	Homogeneous	16-D QoS Vector	429,992	Yes	GAT-QoS with the three QoS node columns zeroed
HGT	Native Multigraph	Heterogeneous	Relation 1-hot; $w(e)=1$	434,620	Yes	Isolates relational typing without QoS
HGT-QoS	Native Multigraph	Heterogeneous	16-D QoS Vector	434,620	Yes	SaG learned engine
HGT-QoS-U	Native Multigraph	Heterogeneous	16-D QoS Vector	330,895	Yes	Directionality control: no reverse pass
Hybrid-HGT	Native Multigraph	Heterogeneous	16-D QoS Vector	434,941	Yes	HGT-QoS correcting the closed-form engine
Hybrid-GAT	Native Multigraph	Homogeneous	16-D QoS Vector	431,433	Yes	GAT-QoS correcting the closed-form engine
GBM-Feat	Per-node features only	None	None	Trees	Yes	Feature-only learner, no graph model
GBM-Feat-QoS	Per-node features only	None	None	Trees	Yes	GBM-Feat with the three QoS node columns
GAT-P	App-Lib <code>DEPENDS_ON</code>	Homogeneous	None	437,496	Yes	GAT on the dependency graph (Amendment 9)
GAT-P-QoS	App-Lib <code>DEPENDS_ON</code>	Homogeneous	16-D QoS Vector	429,992	Yes	GAT-QoS on the dependency graph
GAT-P+InDeg	App-Lib <code>DEPENDS_ON</code>	Homogeneous	16-D QoS Vector	431,433	Yes	Hybrid: GAT-P-QoS correcting InDeg ; supplement only (Amendment 13)
HGT-P-QoS	App-Lib <code>DEPENDS_ON</code>	Heterogeneous	16-D QoS Vector	430,680	Yes	HGT-QoS on the dependency graph (width 100)
GBM-P-QoS	App-Lib <code>DEPENDS_ON</code>	None	Node + QoS features	Trees	Yes	Learned I_{dyn} approximation / tabular learner on dependency features
RM / $Q(v)$	G_{analysis} (Analysis Graph)	Per-type rules	Scalar $w(e)$	0	No	Diagnostic attribution reference

Earlier names. The registered analysis plan (`docs/research/jss/PREREGISTRATION.md`) and the published result artifacts use the labels of an earlier naming scheme. Table S33 maps them; `saag/evaluation/variant_registry.py` holds the same map (`LEGACY_LABELS`), and the internal variant identifiers that key every artifact are unchanged.

Substrates. Under LOSO the learned predictors ingest the complete native typed multigraph across all five entity types, except the **-P** learners of Amendment 9, which ingest the Application–Library `DEPENDS_ON`

Table S33: Current predictor names and the earlier labels used by the registered plan and the result artifacts.

Current name	Earlier label	Variant identifier
GAT-S	GAT-N	gl_full (gl under LOSO)
GAT-S-w	GAT-N-QoS	gl_full_qos (gl_qos under LOSO)
GAT-S-P / GAT-S-P-w	GAT / GAT-QoS (in-distribution only)	gl / gl_qos
GAT	GAT-N-C	gl_full_cap
GAT-w	GAT-N-QoS-C	gl_full_qos_cap
GAT-QoS	GAT-N-QoS16-C	gl_full_qos16_cap
Hybrid-HGT	SaG-Hybrid	hgl_qos_prior
Hybrid-GAT	SaG-Hybrid-GAT	gl_qos16_prior

graph with bit-identical node features and labels. In distribution (Section S20), **GAT-S-P/GAT-S-P-w** consume the Application–Library **DEPENDS_ON** projection, which confounds typing with multi-entity visibility, so no comparison is drawn between those families. **GAT-S-w** uses per-type projection with untyped **GATConv** across edges and a scalar $w(e)$; **HGT-QoS** uses relation-specific **HGTConv** weights and all 16 edge dimensions. Besides parameter budget (434,620 vs. 28,168), message directionality also differs (103,725 reverse parameters). In QoS-off arms every edge weight is set to 1 and the QoS node columns are zeroed, but four node centralities are computed on the QoS-weighted projection, so “QoS-off” means “no explicit QoS channel”, not “no QoS information”. Amendment 2 registered four control arms (**GAT**, **GAT-w**, **GAT-QoS**, **HGT-QoS-U**); the two that were run form, with **HGT** and **HGT-QoS**, the matched 2×2 of the main manuscript (**make -f reproduce/Makefile rq2-matched**).

Closed-form projection. The closed-form scores are computed on the Application–Library projection, which carries Rule 1 (a subscriber joined to each publisher of a topic it consumes, at $w = 1 - \prod_{t \in T} (1 - w(t))$) and Rule 5 (an application joined to each library it uses). On this projection the Rule 5 edge carries a neutral constant, the graph’s median topic weight, rather than the harmonic coupling of code-quality weights, so that the closed-form baselines consume no code metrics. The 0.6/0.4 split between the betweenness and articulation terms is a declared convention and is not tuned; Section S25 shows the effect of the articulation term.

Zero-shot configuration sensitivity. An earlier zero-shot configuration used 2 layers and 150 epochs, chosen to limit over-smoothing across the smaller diameters of the system models ($|V_{\text{app}}| \leq 41$). Because that reasoning appeals to a property of the evaluation targets, it is not reported as the primary result. It is uniformly slightly stronger ($\rho = 0.792$ against 0.760; $\rho_{>0} = +0.281$ against +0.236; $\text{Overlap}@K = 0.533$ against 0.470) and leaves every qualitative conclusion unchanged.

S36. Registered LOSO Sweep

The analysis plan registered its two confirmatory contrasts, **HGT-QoS** and **HGT** against **Topo-QoS**, on the GPU sweep of Table S34. The main manuscript reports the CPU runs of the same protocol instead (Table 6 of the main manuscript). Those runs are the ones the hybrid and matched-control registrations were evaluated on, and their comparator rows are bit-identical across the CPU sweeps. The two sweeps agree on every conclusion: **HGT-QoS** reaches 0.638 here and 0.622 on CPU, within the run-to-run displacement of Section 7.4 of the main manuscript. Table S34 also carries the small GATs, whose comparison with **HGT** is the unmatched 2×2 of Section S31, and the explanation layer’s ranking correlation, which is listed for reference only.

Registered contrasts. **HGT-QoS** wins 9 of 12 folds against **Topo-QoS** (+0.085, CI $[-0.029, +0.194]$), but the registered confirmatory contrast is not significant ($p = 0.151$, Holm 0.303; **HGT**: -0.002 , $p = 0.470$). On $\text{Overlap}@K$ the learned engines lead (0.424–0.431 against 0.388), but the fold-level differences are not significant either ($\Delta = +0.037$, $p = 0.470$ for **HGT-QoS**). With the median test–retest correlation of the labels at 0.982, **HGT-QoS** recovers roughly 65% of the attainable signal in this sweep.

Table S34: The registered LOSO sweep (GPU), Application population, twelve folds. Learned variants share training set, the native multigraph, depth and selection rule; **GAT-S/GAT-S-w** are the small (28k-parameter) GATs (Section S35). Fold score = mean over five seeds; **Fold** σ = spread of the twelve fold means; **Seed** σ = median within-fold spread (zero for deterministic baselines); CI = bootstrap over folds ($B = 2,000$). $\Delta\rho$ is paired by fold against the registered comparator **Topo-QoS**, with its own bootstrap interval; for every learned variant the interval spans zero.

Predictor / Reference	Mean LOSO ρ	95% CI	$\Delta\rho$ vs Topo-QoS	Fold σ	Seed σ	Overlap@K	Requires Training
<i>Training-free structural baselines</i>							
Topo	0.349	[0.254, 0.452]	-0.204 [-0.286, -0.122]	0.173	—	0.366	No
Topo-QoS	0.553	[0.443, 0.657]	— (reference)	0.192	—	0.388	No
<i>Learned predictors (shared native substrate, matched training set)</i>							
GAT-S	0.317	[0.254, 0.381]	-0.236 [-0.342, -0.125]	0.111	0.298	0.328	Yes
GAT-S-w	0.604	[0.538, 0.665]	+0.051 [-0.067, +0.169]	0.112	0.024	0.431	Yes
HGT	0.551	[0.474, 0.617]	-0.002 [-0.066, +0.072]	0.124	0.114	0.427	Yes
HGT-QoS	0.638	[0.561, 0.710]	+0.085 [-0.029, +0.194]	0.133	0.052	0.424	Yes
<i>Diagnostic reference — not a ranking model</i>							
RM / Q(v)	0.205	[0.092, 0.320]	-0.348 [-0.432, -0.265]	0.195	—	0.322	No

Where the engines differ. In this sweep HGT-QoS loses substantively on Enterprise (0.461 vs. 0.795) and Telecom RAN (0.407 vs. 0.576), and wins by +0.229 and +0.210 on the two folds where Topo-QoS is weakest (Microservices, ATM). Enterprise is the largest graph (520 nodes), with by far the densest derived projection (26,276 edges). However, neither graph size ($\rho = -0.434$ with the margin, $p = 0.159$) nor density predicts in advance which engine wins on an unseen architecture.

S37. Engine Regimes

This section backs Section 7.2 of the main manuscript. It is post hoc and exploratory and trains nothing: `reproduce/engine_regimes.py` (`make -f reproduce/Makefile jss-regimes`) reads the per-fold and zero-shot artifacts of the registered sweeps, joins them with descriptors read from the committed corpus, and writes `results/engine_regimes.json`. Every arm is read from the sweep that reports it, so each column mean equals the mean printed in the main manuscript. **GBM-Feat** is gradient-boosted trees over each Application’s own feature vector (Section 3.4 of the main manuscript) with no graph model; **GBM-Feat-QoS** adds the three QoS node columns.

Regimes. Folds are grouped into terciles of Topo-QoS accuracy: W (closed-form ranks poorly), I (intermediate) and S (closed-form ranks well). The grouping uses the labels, so it describes the corpus and cannot place an unseen architecture. Because it conditions on the comparator, part of the per-tercile *gain* of any engine over Topo-QoS is regression to the mean; the main manuscript therefore quotes each engine’s own per-tercile ρ rather than its gain. Table S35 synthesizes the empirical evidence across the five operational regimes.

Descriptors. Table S37 lists the twelve descriptors correlated against the per-fold gains: entity counts, edges per node, **USES** edges per application, the Gini coefficient of subscribers per topic, the inert share ($I^* = 0$), the share of tied labels, the density of the Application **DEPENDS_ON** projection, and the number of distinct topic QoS profiles (not shown; it tracks topic count, because the deadline is continuous). The last column is the share of the graph from which HGT-QoS reaches an Application (`results/receptive_field_probe.json`); for every GAT arm and for HGT-QoS-U it is the Application itself.

Correlations. Table S38 gives Spearman’s ρ between each descriptor and each per-fold quantity over the twelve folds, 96 correlations in all. Benjamini–Hochberg q-values are computed across the whole matrix; the smallest is 0.32, so no cell survives, and the table is a source of hypotheses only. HGT-QoS’s receptive-field share is equally uninformative: its Spearman correlation is +0.12 with HGT-QoS’s per-fold ρ , -0.06 with its gain over Topo-QoS, and +0.20 with Hybrid-HGT’s gain over HGT-QoS.

Zero-shot transfer. Table S39 extends Table 9 of the main manuscript to every arm. The pure learned engines lead on the three models of originally publish–subscribe systems in both ρ and $\rho_{>0}$. On the two RPC-derived models their full-population ρ stays high, but no engine orders the active components reliably.

Table S35: Where graph learning helps. LOSO folds are grouped into terciles of closed-form accuracy (Topo-QoS ρ); system models are grouped by architectural paradigm. Mean Spearman ρ against $I^*(v)$, Application population; $\rho_{>0}$: active stratum. For system models, “learned” ranges over the pure learned engines and “baseline” over Topo and Topo-QoS. The last column gives the reference counts (Amendment 7), which restate I^* ’s rule and are not predictors (Amendment 13), and the dependency-graph learner (Amendment 9). “Best choice” ranges over predictors only. Exploratory. Because the terciles are formed on the comparator’s own score, a learner’s gain over Topo-QoS is inflated in the lowest tercile and deflated in the highest by regression to the mean; with four folds per tercile the pattern is descriptive only.

Regime	Where	Best choice	Evidence: raw-multigraph engines	Dependency graph (references; learner)
Closed-form ranks poorly	Microservices, ATM, IoT Smart City, Healthcare	Dependency-graph learner	Topo-QoS 0.324; HGT-QoS 0.604, GAT-QoS 0.626, both 4/4 folds. The prior trims the gain (hybrids 0.502 / 0.554).	InDeg 0.693, Reach 0.700, GAT-P-QoS 0.716.
Intermediate	ESB, Telecom RAN, Financial Trading, Industrial SCADA	Dependency-graph learner	Topo-QoS 0.561; HGT-QoS 0.589, GAT-QoS 0.660; Hybrid-HGT 0.681, Hybrid-GAT 0.697, both 4/4 folds.	InDeg 0.742, GAT-P-QoS 0.760.
Closed-form ranks well	Logistics Fleet, AV System, Enterprise, Real-Time Gaming	Hybrid engine	Topo-QoS 0.775; HGT-QoS 0.672 (1/4 folds), GAT-QoS 0.620 (0/4); hybrids 0.786 / 0.798.	InDeg 0.858, GAT-P-QoS 0.768.
Unlike the corpus, originally pub-sub	Autoware, EdgeX, Home Assistant	Learned engine	Learned 0.716–0.927 vs. baseline 0.289–0.534; learned $\rho_{>0}$ 0.183–0.833 where baseline is negative.	Reach 0.836–0.997; $\rho_{>0}$ 0.674–0.971.
Unlike the corpus, originally RPC	Online Boutique, Train-Ticket models	Mixed	Learned 0.710–0.810, but application-layer baseline 0.891 on Online Boutique (Supplementary Section S44); learned $\rho_{>0}$ –0.19 to +0.16.	Reach 0.966–0.998; $\rho_{>0}$ 0.813–0.976.

Table S36: Per-fold LOSO Spearman ρ of every arm, Application population, mean over five seeds. T: tercile of Topo-QoS accuracy. Folds are ordered by Topo-QoS.

Fold	T	Topo-QoS	GAT	GAT-QoS	GAT-w	HGT	HGT-QoS	HGT-QoS-U	GBM-Feat	GBM-Feat-QoS	Hybrid-HGT	Hybrid-GAT
Microservices	W	0.265	0.326	0.479	0.442	0.345	0.475	0.528	0.427	0.453	0.366	0.429
ATM	W	0.311	0.398	0.506	0.482	0.329	0.523	0.493	0.304	0.319	0.429	0.447
IoT Smart City	W	0.351	0.609	0.720	0.716	0.580	0.688	0.616	0.721	0.691	0.590	0.654
Healthcare	W	0.369	0.716	0.798	0.803	0.585	0.730	0.816	0.781	0.709	0.625	0.686
ESB (Hub-and-Spoke)	I	0.430	0.500	0.630	0.631	0.335	0.548	0.633	0.539	0.553	0.564	0.568
Telecom RAN	I	0.576	0.520	0.574	0.591	0.576	0.427	0.639	0.703	0.700	0.648	0.656
Financial Trading	I	0.586	0.721	0.713	0.718	0.626	0.695	0.720	0.762	0.753	0.754	0.797
Industrial SCADA	I	0.650	0.557	0.721	0.730	0.531	0.684	0.647	0.856	0.855	0.758	0.768
Logistics Fleet	S	0.741	0.575	0.654	0.651	0.673	0.771	0.668	0.622	0.634	0.792	0.806
AV System	S	0.753	0.638	0.732	0.731	0.619	0.704	0.734	0.748	0.764	0.782	0.793
Enterprise	S	0.795	0.511	0.407	0.411	0.633	0.426	0.411	0.533	0.475	0.735	0.768
Real-Time Gaming	S	0.810	0.683	0.685	0.687	0.751	0.789	0.674	0.710	0.678	0.837	0.825
Mean		0.553	0.563	0.635	0.633	0.548	0.622	0.632	0.642	0.632	0.657	0.683
Mean $\rho_{>0}$		0.280	0.256	0.338	0.338	0.265	0.335	0.355	0.391	0.384	0.340	0.398

Table S37: Per-fold descriptors and HGT-QoS receptive-field share (RF).

Fold	$ V_{app} $	Topics	Brokers	Hosts	Libs	$ E / V $	USES/app	Sub. Gini	Inert	Ties	Proj. dens.	RF
Microservices	90	45	6	15	30	3.65	0.94	0.22	0.14	0.29	0.072	0.39
ATM	26	27	5	8	8	3.53	1.42	0.25	0.19	0.27	0.182	0.48
IoT Smart City	200	80	6	30	10	3.64	0.37	0.31	0.27	0.37	0.032	0.35
Healthcare	50	25	3	8	12	3.97	1.60	0.32	0.36	0.60	0.070	0.56
ESB (Hub-and-Spoke)	70	30	2	12	25	4.97	2.90	0.20	0.23	0.63	0.067	0.59
Telecom RAN	120	55	8	20	22	3.92	0.96	0.30	0.17	0.28	0.057	0.40
Financial Trading	60	35	5	6	18	5.09	1.42	0.23	0.45	0.60	0.127	0.59
Industrial SCADA	140	70	4	25	15	3.24	0.78	0.37	0.38	0.50	0.018	0.39
Logistics Fleet	110	50	7	18	20	3.68	0.97	0.28	0.29	0.41	0.056	0.40
AV System	80	40	4	8	20	5.09	1.59	0.22	0.49	0.74	0.105	0.58
Enterprise	300	120	10	40	50	6.18	1.61	0.20	0.39	0.51	0.054	0.48
Real-Time Gaming	75	38	5	12	28	3.99	3.23	0.32	0.36	0.63	0.032	0.44

Table S38: Spearman correlation of per-fold quantities with fold descriptors ($n = 12$). Asterisk: nominal $p < 0.05$; no cell survives Benjamini–Hochberg correction across the matrix.

Descriptor	HGT-QoS – Topo-QoS	GAT-QoS – Topo-QoS	GBM-Feat – Topo-QoS	HGT – GAT	HGT-QoS – GAT-QoS	Hybrid-HGT – HGT-QoS	GAT-QoS – GAT	GAT-QoS – GBM-Feat-QoS
[Vapp]	−0.48	−0.36	−0.10	+0.63*	+0.03	+0.43	+0.00	−0.60*
Topics	−0.50	−0.41	−0.14	+0.66*	+0.10	+0.46	−0.04	−0.64*
Brokers	−0.45	−0.43	−0.41	+0.77*	+0.42	+0.29	−0.46	−0.35
Hosts	−0.32	−0.25	−0.12	+0.56	+0.01	+0.25	+0.10	−0.40
Libs	−0.62*	−0.51	−0.52	+0.57	+0.23	+0.33	−0.29	−0.36
E / V	−0.50	−0.42	−0.37	+0.12	+0.08	+0.48	−0.66*	−0.21
USES/app	−0.24	−0.34	−0.50	−0.03	+0.20	+0.17	−0.46	+0.21
QoS profiles	−0.53	−0.43	−0.12	+0.66*	+0.08	+0.50	−0.03	−0.70*
Sub. Gini	+0.25	+0.15	+0.45	−0.03	−0.13	−0.17	+0.15	−0.08
Inert	−0.35	−0.42	−0.07	+0.05	+0.18	+0.53	−0.40	−0.43
Ties	−0.20	−0.20	−0.07	−0.20	−0.06	+0.22	−0.18	−0.08
Proj. dens.	+0.24	+0.27	−0.02	−0.41	+0.01	−0.22	−0.07	+0.40

Table S39: Zero-shot Spearman ρ (top) and active-stratum $\rho_{>0}$ (bottom) against $I^*(v)$ on the five system models, Application population. Learned arms are trained on all twelve synthetic scenarios (3 layers, 300 epochs, five seeds).

System model	Topo	Topo-QoS	GAT	GAT-QoS	GAT-w	HGT-QoS	HGT-QoS-U	GBM-Feat	GBM-Feat-QoS	Hybrid-HGT	Hybrid-GAT
ρ											
Autware (ROS 2)	0.307	0.378	0.778	0.758	0.744	0.716	0.755	0.698	0.694	0.596	0.576
EdgeX Foundry	0.534	0.534	0.853	0.815	0.827	0.793	0.807	0.792	0.781	0.702	0.748
Home Assistant	0.297	0.289	0.927	0.925	0.898	0.864	0.921	0.736	0.685	0.725	0.748
Online Boutique (model)	0.891	0.888	0.810	0.750	0.744	0.710	0.765	0.747	0.822	0.747	0.595
Train-Ticket (model)	0.528	0.541	0.786	0.777	0.757	0.717	0.773	0.810	0.812	0.706	0.642
Mean	0.511	0.526	0.831	0.805	0.794	0.760	0.804	0.757	0.759	0.695	0.662
$\rho_{>0}$											
Autware (ROS 2)	−0.100	−0.073	0.676	0.629	0.605	0.517	0.600	0.479	0.527	0.337	0.359
EdgeX Foundry	−0.399	−0.399	0.372	0.255	0.233	0.183	0.326	−0.255	−0.348	−0.371	0.032
Home Assistant	−0.123	−0.143	0.829	0.833	0.749	0.702	0.828	0.312	0.190	0.405	0.358
Online Boutique (model)	−0.024	−0.072	0.055	0.024	−0.043	−0.031	0.156	0.122	0.228	0.069	−0.261
Train-Ticket (model)	0.228	0.228	−0.045	−0.148	−0.077	−0.192	−0.154	0.120	0.248	0.082	0.103
Mean	−0.083	−0.092	0.377	0.319	0.293	0.236	0.351	0.156	0.169	0.105	0.118

S38. Amendment 7: Dependency Counts and QoS-Attribution Controls

Amendment 7 of the registered plan ([docs/research/jss/PREREGISTRATION.md](#)) fixed, before any of its numbers existed, four training-free rankers on the Application–Library projection, three controls on where the closed-form gain comes from, a sweep of the oracle’s free parameters and an inert-vs-active rule, with decision rules for each outcome. Every arm was run and is reported here. Amendment 13 later reclassified the dependency counts as references that restate I^* ’s propagation rule, so the main manuscript no longer reports them as predictors; the values here are unchanged. The harness ([reproduce/training_free_suite.py](#)) needs neither PyTorch nor Neo4j; it regenerates $I^*(v)$ with the published oracle settings and first reproduces every published per-fold Topo–QoS value to three decimals ([results/tf_reproduction_gate.json](#)). The tables below are rendered from the artifacts by [reproduce/render_amendment7_tables.py](#).

Outcomes of the decision rules. R1 applies: the best new ranker (InDeg, 0.764) exceeds HGT–QoS (0.622), so the main manuscript states that the learned engines do not beat a training-free dependency count on this oracle (after Amendment 13, a reference count); the learners on the dependency graph of Amendment 9 match it. R2 applies: a constant topic weight (Topo-Mult, 0.595) and permuted QoS profiles (Perm, 0.559) keep the whole gain of Topo–QoS (0.553) over the registered Topo (0.349), so that gain is attributed to QoS-weighted dependency multiplicity on the projection rather than to the content of the declared QoS contracts. R2’ applies as well: on the QoS-independent corpus, QoS weighting changes projection betweenness by -0.035 , so the generator’s QoS–topology coupling is part of the mechanism. The registered Topo read betweenness from the analysis stage’s application-layer graph; rebuilt in memory, that score reproduces the published column to within 0.011 on average (0.360 vs. 0.349; Table S43). R3 applies: every arm is reported here.

A reproducibility finding. The CDI’s breadth-first sample is the highest-degree nodes of a Python set, so equal-degree ties are ordered by string hashing, which Python salts per process. Unpinned, the CDI arm moved between runs (mean ρ 0.239 vs. 0.253). The harness pins PYTHONHASHSEED=0. Because CDI is also a node feature of the learned engines and of the explanation layer, this source of run-to-run variation applies to them too, and is one candidate contributor to the learned-cell drift of Section 7.4 of the main manuscript.

S39. Amendments 9 and 10: Learning on the Dependency Graph and the Value of the Derivation

Amendment 9 moved SaG’s learners, at their registered parameter budgets and with bit-identical node features and labels, from the raw multigraph onto the Application–Library `DEPENDS_ON` graph, and registered

Table S40: Training-free rankers of Amendment 7, per LOSO holdout: Spearman ρ against $I^*(v)$, Application population, labels regenerated with the published oracle settings. **Topo-QoS** reproduces the published per-fold values to three decimals (**results/tf_reproduction_gate.json**). Betweenness (proj.) is unweighted betweenness on the same Application–Library projection. Rendered from **results/tf_baselines.json**.

Holdout	Topo-QoS	Betweenness (proj.)	InDeg	Reach	Reach-QoS	CDI
ATM	0.311	0.291	0.495	0.670	0.389	−0.359
AV System	0.753	0.791	0.868	0.835	0.840	0.164
Enterprise	0.795	0.844	0.891	0.826	0.836	0.259
Financial Trading	0.586	0.694	0.874	0.803	0.824	0.193
Healthcare	0.369	0.402	0.831	0.819	0.783	0.280
Enterprise Integration (ESB)	0.430	0.544	0.529	0.612	0.486	−0.104
Industrial SCADA	0.650	0.668	0.868	0.776	0.817	0.468
IoT Smart City	0.351	0.367	0.898	0.655	0.763	0.482
Logistics Fleet	0.741	0.736	0.836	0.759	0.806	0.514
Microservices	0.265	0.348	0.548	0.656	0.436	0.188
Real-Time Gaming	0.810	0.855	0.836	0.834	0.850	0.266
Telecom RAN	0.576	0.556	0.698	0.535	0.736	0.450
Mean ρ	0.553	0.591	0.764	0.732	0.714	0.233
Mean $\rho_{>0}$ (active)	0.280	0.294	0.516	0.286	0.399	0.274
Mean Overlap@K	0.388	0.411	0.504	0.344	0.465	0.342

Table S41: Paired contrasts of the training-free rankers over the twelve LOSO folds (two-sided Wilcoxon, bootstrap 95% CI of the mean difference, $B = 2,000$). The registered family (Amendment 7) is the four new rankers against **Topo-QoS**, Holm-corrected in parentheses; the unweighted projection betweenness is descriptive. The last five columns pair each ranker with the published CPU per-fold values of the learned and hybrid engines (Section S26) and of the feature-only regressor (Amendment 8, Section S37): $\Delta\rho$ (folds won, p).

Ranker	$\Delta\rho$ vs Topo-QoS [95% CI]	Won	p (PTolm)	vs HGT-QoS	vs GAT-QoS	vs Hybrid-HGT	vs Hybrid-GAT	vs GBM-Feat
InDeg	+0.211 [+0.132, +0.299]	12/12	0.0005 (0.0020)	+0.143 (10/12, 0.0024)	+0.129 (10/12, 0.0049)	+0.108 (10/12, 0.0024)	+0.081 (11/12, 0.0024)	+0.122 (10/12, 0.0024)
Reach	+0.178 [+0.088, +0.268]	11/12	0.0034 (0.0068)	+0.110 (10/12, 0.0024)	+0.097 (9/12, 0.0161)	+0.075 (9/12, 0.0425)	+0.049 (10/12, 0.0771)	+0.089 (9/12, 0.0923)
Reach-QoS	+0.161 [+0.097, +0.235]	12/12	0.0005 (0.0020)	+0.092 (9/12, 0.0771)	+0.079 (8/12, 0.1099)	+0.057 (10/12, 0.0269)	+0.031 (9/12, 0.1099)	+0.072 (10/12, 0.0210)
CDI	−0.320 [−0.451, −0.189]	1/12	0.0034 (0.0068)	−0.388 (1/12, 0.0010)	−0.402 (0/12, 0.0005)	−0.423 (0/12, 0.0005)	−0.450 (0/12, 0.0005)	−0.409 (0/12, 0.0005)
Betweenness (proj.)	+0.038 [+0.014, +0.063]	9/12	0.0210	−0.030 (4/12, 0.5693)	−0.044 (4/12, 0.4697)	−0.065 (3/12, 0.0425)	−0.092 (2/12, 0.0122)	−0.051 (5/12, 0.3804)

Table S42: Training-free rankers on the five hand-authored system models: Spearman ρ / Overlap@K, Application population. Same oracle settings as Table 9 of the main manuscript, but projection and closed-form scores are computed by the Amendment 7 harness from the committed topology files; **Topo-QoS** scores 0.582 here against 0.526 in the main manuscript’s zero-shot table, whose reference scores use the native-graph projection path and the cached articulation term. Rendered from **results/tf_baselines.json**.

System model	$ V_{app} $	Topo-QoS	Betweenness (proj.)	InDeg	Reach	Reach-QoS	CDI
Autoware.universe (ROS 2)	32	0.462 / 0.333	0.464 / 0.333	0.620 / 0.500	0.836 / 0.500	0.881 / 0.667	0.193 / 0.167
EdgeX Foundry	22	0.621 / 0.000	0.621 / 0.000	0.896 / 0.250	0.997 / 1.000	0.934 / 0.500	0.550 / 0.250
Home Assistant	24	0.383 / 0.000	0.449 / 0.000	0.943 / 0.600	0.891 / 0.800	0.933 / 0.800	0.358 / 0.400
Online Boutique (pub-sub model)	22	0.909 / 0.500	0.951 / 0.500	0.988 / 0.750	0.998 / 0.750	0.995 / 1.000	0.818 / 0.250
Train-Ticket	41	0.535 / 0.625	0.598 / 0.750	0.867 / 0.500	0.966 / 0.750	0.921 / 0.625	0.375 / 0.500
Mean	—	0.582 / 0.292	0.617 / 0.317	0.863 / 0.520	0.938 / 0.760	0.933 / 0.718	0.459 / 0.313
Mean PR-AUC	—	0.518	0.526	0.752	0.933	0.888	0.430
Mean $\rho_{>0}$ (active)	—	−0.006	−0.016	0.321	0.871	0.670	−0.106

Table S43: Where the closed-form gain comes from, per LOSO holdout (Spearman ρ , Application population). *Published Topo*: the registered comparator, which read betweenness from the analysis stage’s application-layer dependency graph; *app-layer BT*: that betweenness rebuilt in memory (`MemoryRepository`, which differs slightly from Neo4j on Rule-1 weights). *Proj. BT*: unweighted betweenness on the Application–Library projection *Topo-QoS* uses. *Topo-Mult*: *Topo-QoS* with every topic weight fixed at 0.5. *Perm*: *Topo-QoS* with QoS profiles permuted across topics (mean of 20). *Indep.*: the corpus regenerated with QoS no longer steering topology (`qos_affinity: false`), relabeled. Rendered from `results/qos_attribution_controls.json`, `results/topo_substrate_check.json` and `results/qos_indep_corpus.json`.

Holdout	Substrate			QoS content			Indep. corpus	
	Published Topo	App-layer BT	Proj. BT	Topo-QoS	Topo-Mult	Perm	Proj. BT	Topo-QoS
ATM	0.256	0.256	0.291	0.311	0.350	0.285	0.323	0.337
AV System	0.468	0.505	0.791	0.753	0.784	0.766	0.757	0.753
Enterprise	0.431	0.448	0.844	0.795	0.840	0.801	0.851	0.831
Financial Trading	0.237	0.240	0.694	0.586	0.702	0.641	0.773	0.695
Healthcare	0.077	0.101	0.402	0.369	0.374	0.381	0.598	0.524
Enterprise Integration (ESB)	0.112	0.125	0.544	0.430	0.571	0.398	0.737	0.578
Industrial SCADA	0.608	0.607	0.668	0.650	0.666	0.668	0.544	0.564
IoT Smart City	0.251	0.251	0.367	0.351	0.371	0.382	0.318	0.325
Logistics Fleet	0.576	0.580	0.736	0.741	0.733	0.720	0.646	0.637
Microservices	0.229	0.231	0.348	0.265	0.346	0.338	0.348	0.265
Real-Time Gaming	0.377	0.407	0.855	0.810	0.854	0.802	0.642	0.632
Telecom RAN	0.562	0.569	0.556	0.576	0.552	0.523	0.646	0.622
Mean	0.349	0.360	0.591	0.553	0.595	0.559	0.598	0.564

Table S44: Sensitivity of $I^*(v)$ to its propagation threshold θ and per-wave damping step (floor 0.25). Label agreement is Spearman ρ against the shipped setting ($\theta = 0.2$, step 0.15) on Applications, mean and minimum over the twelve folds; the last column is *Topo-QoS* scored against each label set. Rendered from `results/oracle_param_sensitivity.json`.

θ	Damping step	Agreement (mean)	Agreement (min)	Topo-QoS ρ
0.1	0.10	0.879	0.656	0.500
0.1	0.15	0.885	0.720	0.506
0.1	0.20	0.862	0.671	0.501
0.2	0.10	0.986	0.913	0.557
0.2	0.15	1.000	1.000	0.553
0.2	0.20	0.984	0.888	0.558
0.3	0.10	0.942	0.791	0.578
0.3	0.15	0.942	0.790	0.577
0.3	0.20	0.942	0.790	0.577

Table S45: Label structure of the LOSO folds and the system models (means), and the inert-vs-active rule of Amendment 7: a component is predicted to propagate failure ($I^* > 0$) iff it has at least one transitive dependent on the projection. Rendered from `results/system_model_descriptives.json` and `results/tf_baselines.json`.

Set	$ V_{\text{app}} $	Zero share	Tie fraction	Proj. density	Rule bal. acc.
Twelve LOSO folds	110	0.309	0.485	0.073	0.940
Five system models	28	0.508	0.625	0.050	0.973

twelve contrasts against **InDeg**, **Reach** and each learner’s raw-multigraph counterpart before any of the learned arms had run (`make -f reproduce/Makefile rq-dependency-graph`; one clean CPU invocation, re-run from a clean tree with bit-identical results). The re-run **Topo-QoS** and **GAT-QoS** reproduce their earlier per-fold values exactly, and **InDeg** and **Reach**, recomputed on the LOSO labels, match Amendment 7’s artifact exactly. **HGT-P-QoS** did not train stably (median within-fold seed spread 0.208, against 0.026 for **GAT-P-QoS**); it is reported as registered, without tuning.

Under Amendment 13, **InDeg**, **Reach**, **Pubs-raw** and **Reach-R1** are references, not predictors, and **GAT-P+InDeg**, whose prior is **InDeg**, is reported only here (Table S46: $\rho = 0.758$, a correction to the reference count that adds nothing measurable to it). The contrasts below are descriptive comparisons with the references.

Amendment 10 answers a referee’s question: what does the derivation contribute beyond a subscriber count? For an Application, **InDeg** is the raw 2-hop subscriber count, and removing Rule 5 cannot change it, because Rule-5 edges point into Libraries; both identities are pinned by tests on all seventeen graphs. The registered question is therefore whether the derived count beats the counts available on the raw multigraph without derivation, and whether the derived library rule adds to transitive reach. Both decision rules apply (Table S50).

Table S46: Learners on the dependency graph (Amendment 9) beside their raw-multigraph counterparts and the dependency counts, per LOSO holdout: Spearman ρ against $I^*(v)$, Application population, mean over five seeds. Raw-multigraph values are read from their own clean artifacts; **InDeg** and **Reach** are recomputed on the same labels and match Table S40 exactly. Rendered from `results/dependency_graph_contrasts.json`.

Holdout	InDeg	Reach	GAT	GAT-P	GAT-QoS	GAT-P-QoS	Hybrid-GAT	GAT-P+InDeg	HGT-QoS	HGT-P-QoS
ATM	0.495	0.670	0.398	0.406	0.506	0.603	0.447	0.501	0.523	0.181
AV System	0.868	0.835	0.638	0.736	0.732	0.764	0.793	0.832	0.704	0.679
Enterprise	0.891	0.826	0.511	0.777	0.407	0.738	0.768	0.872	0.426	0.758
Financial Trading	0.874	0.803	0.721	0.767	0.713	0.800	0.797	0.848	0.695	0.734
Healthcare	0.831	0.819	0.716	0.753	0.798	0.863	0.686	0.823	0.730	0.783
Enterprise Integration (ESB)	0.529	0.612	0.500	0.519	0.630	0.758	0.568	0.525	0.548	0.464
Industrial SCADA	0.868	0.776	0.557	0.698	0.721	0.834	0.768	0.863	0.684	0.254
IoT Smart City	0.898	0.655	0.609	0.590	0.720	0.747	0.654	0.891	0.688	0.209
Logistics Fleet	0.836	0.759	0.575	0.622	0.654	0.774	0.806	0.827	0.771	0.546
Microservices	0.548	0.656	0.326	0.611	0.479	0.649	0.429	0.544	0.475	0.547
Real-Time Gaming	0.836	0.834	0.683	0.797	0.685	0.795	0.825	0.851	0.789	0.601
Telecom RAN	0.698	0.535	0.520	0.556	0.574	0.647	0.656	0.718	0.427	0.416
Mean	0.764	0.732	0.563	0.653	0.635	0.748	0.683	0.758	0.622	0.514
Mean $\rho_{>0}$	0.516	0.286	0.256	0.336	0.338	0.440	0.398	0.537	0.335	0.237

S40. Amendment 12: Robustness Analyses

This section backs the round-7 revision of the main manuscript. Everything in it is exploratory. Under Amendment 13, the R1 contrasts compare the **InDeg** reference with raw-graph rankers and carry no claim in the main text, and the learned rows include **GAT-P+InDeg**, which the main text omits because its prior is a reference. `reproduce/referee_round7.py` (`make -f reproduce/Makefile rq-referee-round7`) computes the analyses registered in Amendment 12, and `reproduce/render_referee_tables.py` renders the tables below from its artifacts. Four deviations from the registration are recorded in `docs/research/jss/PREREGISTRATION.md`:

- I_{dyn} is the published $n = 30$ lexical sample, because Amendment 11’s full-population labeling had not produced labels when round 7 was run. It has since been completed; the main manuscript now uses the full population, and Section S41 reports this sample as the sensitivity check.
- Gate G3 fails by construction for the learned engines, which are therefore reported as seed ensembles.
- I^* is not timed at 10,000 components.
- A population defect was found and corrected. `subscriber_count_raw` and `pubs_raw` emit only publishers, so non-publishers dropped out of those rankers’ scored population. The defect also affected

Table S47: The twelve registered contrasts of Amendment 9: two-sided Wilcoxon over the twelve folds, fold-bootstrap 95% CI ($B = 2,000$), Holm within this exploratory family. Decision rules: D1 not triggered, D2 not triggered, D3 triggered, M triggered, Z triggered. Rendered from `results/dependency_graph_contrasts.json`.

Contrast	$\Delta\rho$	95% CI	Won	p	p_{Holm}
GAT-P vs InDeg	-0.111	$[-0.163, -0.059]$	1/12	0.0024	0.017
GAT-P vs Reach	-0.079	$[-0.122, -0.046]$	1/12	0.0010	0.009
GAT-P vs GAT	+0.090	$[+0.042, +0.148]$	11/12	0.0015	0.012
GAT-P-QoS vs InDeg	-0.017	$[-0.072, +0.051]$	4/12	0.4697	1.000
GAT-P-QoS vs Reach	+0.016	$[-0.024, +0.058]$	6/12	0.5693	1.000
GAT-P-QoS vs GAT-QoS	+0.113	$[+0.077, +0.162]$	12/12	0.0005	0.006
GAT-P+InDeg vs InDeg	-0.006	$[-0.015, +0.002]$	3/12	0.2036	0.881
GAT-P+InDeg vs Reach	+0.026	$[-0.035, +0.086]$	8/12	0.3804	1.000
GAT-P+InDeg vs Hybrid-GAT	+0.075	$[+0.039, +0.113]$	11/12	0.0034	0.021
HGT-P-QoS vs InDeg	-0.250	$[-0.377, -0.143]$	0/12	0.0005	0.006
HGT-P-QoS vs Reach	-0.217	$[-0.317, -0.131]$	0/12	0.0005	0.006
HGT-P-QoS vs HGT-QoS	-0.107	$[-0.238, +0.014]$	4/12	0.1763	0.881

Table S48: Zero-shot transfer to the five system models (descriptive): learners on the dependency graph trained on all twelve synthetic scenarios (3 layers, 300 epochs, five seeds) beside their raw-multigraph counterparts and the dependency counts. Spearman ρ against $I^*(v)$, Application population. Rendered from `results/dependency_graph_contrasts.json`.

System model	InDeg	Reach	GAT	GAT-P	GAT-QoS	GAT-P-QoS	Hybrid-GAT	GAT-P+InDeg	HGT-QoS	HGT-P-QoS
Autoware.universe (ROS 2)	0.620	0.836	0.778	0.821	0.758	0.794	0.576	0.682	0.716	0.657
EdgeX Foundry	0.896	0.997	0.853	0.846	0.815	0.841	0.748	0.813	0.793	0.786
Home Assistant	0.943	0.891	0.927	0.838	0.925	0.832	0.748	0.905	0.864	0.805
Online Boutique (pub-sub model)	0.988	0.998	0.810	0.829	0.750	0.790	0.595	0.811	0.710	0.758
Train-Ticket	0.867	0.966	0.786	0.815	0.777	0.774	0.642	0.752	0.717	0.724
Mean	0.863	0.938	0.831	0.830	0.805	0.806	0.662	0.792	0.760	0.746

Table S49: Receptive field of the dependency-graph learners at an Application (random weights, 3 layers): mean receptive-field size of the GAT, the share of Applications whose receptive field is exactly the Application and its dependents within three hops, and the share of the graph the bidirectional HGT reaches. Trained zero-shot checkpoints (seed 42), largest change in any Application prediction when every edge is deleted: GAT-P 0.51; GAT-P-QoS 0.54; GAT-P+InDeg 0.32; HGT-P-QoS 0.61. Rendered from `results/dependency_graph_contrasts.json`.

Holdout	$ V_{\text{app}} $	GAT RF (nodes)	RF = 3-hop dependents	HGT RF share
ATM	26	16.0	1.000	0.707
AV System	80	55.5	1.000	1.000
Enterprise	300	206.8	1.000	0.986
Financial Trading	60	45.5	1.000	0.982
Healthcare	50	30.9	1.000	0.917
Enterprise Integration (ESB)	70	75.6	1.000	0.991
Industrial SCADA	140	26.8	1.000	0.510
IoT Smart City	200	63.7	1.000	0.528
Logistics Fleet	110	61.9	1.000	0.861
Microservices	90	74.2	1.000	0.886
Real-Time Gaming	75	41.0	1.000	0.948
Telecom RAN	120	87.7	1.000	0.912

Table S50: The value of the dependency derivation (Amendment 10): dependency counts on the derived graph against counts available on the raw multigraph without derivation (Degree-raw: total degree; Pubs-raw: topics published) and against Rule-1-only reach (Reach-R1). Spearman ρ , Application population. Registered contrasts: InDeg vs Degree-raw +0.565 (12/12, Holm $p = 0.0015$); InDeg vs Pubs-raw +0.033 (11/12, Holm $p = 0.0020$); Reach vs Reach-R1 +0.058 (9/12, Holm $p = 0.0068$). Decision rules: E1 triggered, E2 triggered. For every Application in all seventeen graphs, InDeg equals the raw 2-hop subscriber count ($\max |\Delta| = 0$). Rendered from `results/derivation_ablation.json`.

Holdout	InDeg	Degree-raw	Pubs-raw	Reach	Reach-R1
ATM	0.495	−0.143	0.488	0.670	0.670
AV System	0.868	0.120	0.827	0.835	0.801
Enterprise	0.891	0.356	0.891	0.826	0.662
Financial Trading	0.874	0.111	0.862	0.803	0.757
Healthcare	0.831	−0.054	0.750	0.819	0.792
Enterprise Integration (ESB)	0.529	−0.132	0.432	0.612	0.606
Industrial SCADA	0.868	0.355	0.809	0.776	0.743
IoT Smart City	0.898	0.414	0.879	0.655	0.655
Logistics Fleet	0.836	0.485	0.823	0.759	0.631
Microservices	0.548	0.118	0.523	0.656	0.487
Real-Time Gaming	0.836	0.199	0.800	0.834	0.841
Telecom RAN	0.698	0.565	0.691	0.535	0.441
Mean	0.764	0.199	0.731	0.732	0.674
Five system models	0.863	0.357	0.489	0.938	0.938

Amendment 10’s published Pubs-raw arm, which moves from 0.431 to 0.731; InDeg’s margin over it falls from +0.334 to +0.033, and decision E1 still applies.

Raw-multigraph rankers (R1). Tables S51–S54 report per fold and per oracle. The identity InDeg = Raw2Hop holds on every graph (Remark 1 of the main manuscript), so Raw2Hop is not tabulated. PageRank on the raw graph is constant for every Application, because Applications have no incoming raw edges.

Beyond first order (R2). Table S55 gives the partial correlations per fold. They vary widely between folds (−0.08 to 0.56 for InDeg), as expected on 30 Applications.

Learned engines on every oracle (R3), averaging and seed spread. See Tables S56 and S57. Note in particular the seed spread of the HGT engines: single seeds differ from others in the same fold by more than 1.0 in ρ .

Recall (R4) is plotted in Figure 5 of the main manuscript; the per-fold values are in `data/benchmarks/referee_round7_recall.json`. **System-model descriptors (R5)** are in Table S58, and **latency (R6)** in Table S59.

S41. Amendment 14: Degree-Feature, Aggregator and Queue-Flow Analyses

Everything in this section was registered in Amendment 14 before its arms ran, except the $n = 30$ sensitivity check, which Amendment 11’s rule R requires, and the descriptive tables. Artifacts: `data/benchmarks/referee_round8_*.json`, produced by `reproduce/referee_round8.py`.

Hybrid attribution (F7). Table S60 tests each hybrid against its own base learner (registered in Amendments 5 and 6 and unreported until now), against the registered comparator, and against the two stronger training-free controls of Amendment 7. The sensitivity columns weight folds by $|V_{\text{app}}|$ (exact two-sided sign-flip test over the 2^{12} sign patterns) and apply the Nadeau–Bengio corrected resampled t with test/train ratio 1/11.

Equivalence (F3) and corrected tests. Table S61 gives the distance of the published GAT-P-QoS from the InDeg reference and the Nadeau–Bengio corrected test for the plan’s contrasts. The TOST margin (± 0.05)

Table S51: Raw-multigraph rankers against I^* , per LOSO fold (Amendment 12, R1; Spearman ρ , Application population). PR-row is constant for every Application and is omitted. Rendered from `data/benchmarks/referee_round7_raw_baselines.json`.

Holdout	InDeg	Pubs-row	Degree-row	RevPR-row	Reach-R1	Reach
ATM	0.495	0.488	-0.143	-0.131	0.670	0.670
AV System	0.868	0.827	0.120	0.107	0.801	0.835
Enterprise	0.891	0.891	0.356	0.280	0.662	0.826
Financial Trading	0.874	0.862	0.111	-0.141	0.757	0.803
Healthcare	0.831	0.750	-0.054	-0.172	0.792	0.819
Enterprise Integration (ESB)	0.529	0.432	-0.132	0.079	0.606	0.612
Industrial SCADA	0.868	0.809	0.355	0.155	0.743	0.776
IoT Smart City	0.898	0.879	0.414	0.288	0.655	0.655
Logistics Fleet	0.836	0.823	0.485	0.215	0.631	0.759
Microservices	0.548	0.523	0.118	-0.190	0.487	0.656
Real-Time Gaming	0.836	0.800	0.199	0.207	0.841	0.834
Telecom RAN	0.698	0.691	0.565	0.366	0.441	0.535
Mean	0.764	0.731	0.199	0.089	0.674	0.732

Table S52: Raw-multigraph rankers against I_{dyn} ($n = 30/\text{fold}$), per LOSO fold (Amendment 12, R1; Spearman ρ , Application population). PR-row is constant for every Application and is omitted. Rendered from `data/benchmarks/referee_round7_raw_baselines.json`.

Holdout	InDeg	Pubs-row	Degree-row	RevPR-row	Reach-R1	Reach
ATM	0.358	0.444	-0.082	-0.118	0.447	0.447
AV System	0.761	0.742	0.050	0.172	0.743	0.747
Enterprise	0.731	0.688	0.494	0.274	0.261	0.494
Financial Trading	0.856	0.834	0.251	-0.066	0.585	0.729
Healthcare	0.823	0.691	0.074	-0.149	0.547	0.534
Enterprise Integration (ESB)	0.179	0.091	-0.064	0.224	0.256	0.261
Industrial SCADA	0.357	0.149	0.025	-0.069	0.296	0.380
IoT Smart City	0.886	0.881	0.677	0.580	0.528	0.590
Logistics Fleet	0.790	0.787	0.334	-0.018	0.579	0.708
Microservices	0.587	0.528	0.203	-0.112	0.116	0.359
Real-Time Gaming	0.564	0.558	0.432	0.316	0.546	0.532
Telecom RAN	0.424	0.461	0.407	0.102	0.096	0.274
Mean	0.610	0.571	0.233	0.095	0.417	0.505

Table S53: Raw-multigraph rankers against I_{comp} , per LOSO fold (Amendment 12, R1; Spearman ρ , Application population). PR-raw is constant for every Application and is omitted. Rendered from `data/benchmarks/referee_round7_raw_baselines.json`.

Holdout	InDeg	Pubs-raw	Degree-raw	RevPR-raw	Reach-R1	Reach
ATM	0.646	0.698	0.751	0.610	-0.173	-0.173
AV System	0.721	0.721	0.809	0.492	0.295	0.518
Enterprise	0.736	0.719	0.859	0.584	0.134	0.454
Financial Trading	0.694	0.698	0.858	0.290	0.165	0.461
Healthcare	0.630	0.540	0.625	0.162	0.169	0.422
Enterprise Integration (ESB)	0.501	0.307	0.660	0.127	-0.500	-0.118
Industrial SCADA	0.709	0.605	0.690	0.485	0.365	0.488
IoT Smart City	0.502	0.470	0.806	0.617	0.056	0.051
Logistics Fleet	0.825	0.771	0.778	0.441	0.225	0.502
Microservices	0.685	0.702	0.720	-0.102	-0.065	0.361
Real-Time Gaming	0.476	0.439	0.299	0.159	0.290	0.287
Telecom RAN	0.676	0.724	0.768	0.362	0.002	0.373
Mean	0.650	0.616	0.719	0.352	0.080	0.302

Table S54: The twelve registered R1 contrasts (Amendment 12): InDeg against each untyped or trivially typed raw-graph ranker on each oracle, two-sided Wilcoxon over twelve folds, Holm within the family. PR-raw has no defined ρ , so its contrasts compare InDeg with zero. Identity check: $\max |\text{InDeg} - \text{Raw2Hop}| = 0$ on all seventeen graphs. Rendered from `data/benchmarks/referee_round7_raw_baselines.json`.

Contrast	$\Delta\rho$	95% CI	Won	p	p_{Holm}
I^* : InDeg vs PR-raw	+0.764	[+0.674, +0.840]	12/12	0.0005	0.0059
I^* : InDeg vs RevPR-raw	+0.676	[+0.570, +0.784]	12/12	0.0005	0.0059
I^* : InDeg vs Degree-raw	+0.565	[+0.455, +0.670]	12/12	0.0005	0.0059
I^* : InDeg vs Pubs-raw	+0.033	[+0.017, +0.051]	11/12	0.0010	0.0059
I_{dyn} : InDeg vs PR-raw	+0.610	[+0.475, +0.727]	12/12	0.0005	0.0059
I_{dyn} : InDeg vs RevPR-raw	+0.515	[+0.358, +0.667]	11/12	0.0010	0.0059
I_{dyn} : InDeg vs Degree-raw	+0.376	[+0.255, +0.498]	12/12	0.0005	0.0059
I_{dyn} : InDeg vs Pubs-raw	+0.039	[-0.003, +0.083]	10/12	0.0640	0.1919
I_{comp} : InDeg vs PR-raw	+0.650	[+0.591, +0.706]	12/12	0.0005	0.0059
I_{comp} : InDeg vs RevPR-raw	+0.298	[+0.178, +0.429]	11/12	0.0015	0.0059
I_{comp} : InDeg vs Degree-raw	-0.068	[-0.137, -0.002]	4/12	0.1099	0.2197
I_{comp} : InDeg vs Pubs-raw	+0.034	[-0.002, +0.073]	7/12	0.2036	0.2197

Table S55: Partial Spearman correlation with I_{dyn} after the rank of I^* is regressed out (Amendment 12, R2), per fold, on the published $n = 30$ lexical sample; $\rho(I^*, I_{\text{dyn}})$ is the agreement of the two oracles on the same sample. The last row conditions on the first-order expansion instead of I^* . Rendered from `data/benchmarks/referee_round7_partial.json`.

Holdout	n	$\rho(I^*, I_{\text{dyn}})$	InDeg	Reach	Topo-QoS	First-order
ATM	26	0.691	0.025	−0.031	0.123	0.071
AV System	30	0.787	0.211	0.247	0.182	0.149
Enterprise	30	0.709	0.260	−0.156	0.028	0.263
Financial Trading	30	0.953	0.171	−0.044	−0.141	0.157
Healthcare	30	0.843	0.381	−0.339	0.083	0.309
Enterprise Integration (ESB)	30	0.264	0.029	0.149	0.037	0.068
Industrial SCADA	30	0.186	0.350	0.352	−0.033	0.358
IoT Smart City	30	0.846	0.561	0.042	0.057	0.495
Logistics Fleet	30	0.692	0.528	0.289	0.174	0.388
Microservices	30	0.337	0.518	0.226	0.339	0.651
Real-Time Gaming	30	0.611	0.160	0.150	0.470	0.318
Telecom RAN	30	0.603	−0.081	−0.192	−0.066	0.084
Mean		0.627	0.259	0.058	0.104	0.276
95% CI			[0.143, 0.367]	[−0.059, 0.166]	[0.016, 0.201]	[0.180, 0.375]
Given first order			0.163	0.072	−0.119	—

Table S56: Learned engines re-scored on every oracle (Amendment 12, R3). The saved predictions are the mean of five seeds’ predictions, so registered gate G3 (reproduce the published per-seed mean) fails by construction; the per-seed logs reproduce every published value. Columns: published LOSO ρ ; mean of the per-seed logs; seed-ensemble ρ on each oracle. Rendered from `data/benchmarks/referee_round7_learned_oracles.json`.

Engine	Published	Seed logs	Ensemble I^*	Ensemble I_{dyn}	Ensemble I_{comp}
HGT-QoS	0.622	0.622	0.667	0.440	0.201
GAT-QoS	0.635	0.635	0.647	0.433	0.144
Hybrid-HGT	0.657	0.657	0.672	0.457	0.582
Hybrid-GAT	0.683	0.683	0.702	0.502	0.585
GAT-P-QoS	0.748	0.748	0.772	0.519	0.274
HGT-P-QoS	0.514	0.514	0.618	0.418	0.334
GAT-P+InDeg	0.758	0.758	0.761	0.594	0.578
InDeg	0.764	—	0.764	0.610	0.650

Table S57: Table 6’s means under three averaging rules, and the seed spread of each learned engine: the mean over folds of the SD of the five seeds’ ρ , and the largest within-fold range. Descriptive, computed from existing artifacts. Rendered from `data/benchmarks/referee_round7_averaging.json`.

Ranker	Arithmetic	Fisher- z	$ V_{\text{app}} $ -weighted	Seed SD	Max range
Analytic- I^*	0.808	0.826	0.843	—	—
InDeg	0.764	0.798	0.810	—	—
GAT-P+InDeg	0.758	0.787	0.801	0.011	0.082
GAT-P-QoS	0.748	0.757	0.749	0.030	0.196
Reach	0.732	0.746	0.736	—	—
Hybrid-GAT	0.683	0.703	0.707	0.042	0.316
Hybrid-HGT	0.657	0.678	0.677	0.053	0.415
GAT-QoS	0.635	0.648	0.606	0.024	0.216
HGT-QoS	0.622	0.638	0.595	0.099	1.082
Topo-QoS	0.553	0.586	0.596	—	—
HGT-P-QoS	0.514	0.545	0.518	0.254	1.216
Topo	0.349	0.362	0.389	—	—

Table S58: The five system models beside the synthetic folds (Amendment 12, R5): Applications, share with zero I^* , Gini coefficient of InDeg, longest dependency chain, and InDeg and Reach scored on the learned engines’ own zero-shot labels. Rendered from `data/benchmarks/referee_round7_zeroshot.json`.

System model	$ V_{\text{app}} $	Zero share	Fan-in Gini	Depth	InDeg	Reach
Autoware.universe (ROS 2)	32	0.41	0.54	4	0.620	0.836
EdgeX Foundry	22	0.55	0.68	5	0.896	0.997
Home Assistant	24	0.29	0.53	6	0.943	0.891
Online Boutique (pub-sub model)	22	0.64	0.75	3	0.988	0.998
Train-Ticket	41	0.66	0.77	5	0.867	0.966
<i>Synthetic folds (mean)</i>	110	0.31	0.50	6.0	—	—

Table S59: Counting path against one I^* labeling pass on generated graphs (Amendment 12, R6; median of repeats, count in brackets for I^*). Projection, InDeg and Reach in ms; I^* in s. Rendered from `data/benchmarks/referee_round7_latency.json`.

$ V $	$ E $	Projection (ms)	InDeg (ms)	Reach (ms)	I^* (s)
249	1,081	7.9	0.04	11.7	4.0 (5)
499	2,437	11.1	0.06	24.9	15.1 (5)
999	6,372	25.2	0.09	113.1	58.5 (5)
1,998	19,242	109.6	0.17	490.8	346.6 (5)
4,995	94,790	426.2	0.61	4369.9	2163.5 (1)
9,990	348,277	1094.3	0.95	23302.4	not timed

Table S60: Hybrid contrasts, twelve LOSO folds, I^* , Application population. p_{Holm} : within F7 (four contrasts); the first four rows are Amendments 5/6’s registered contrasts, Holm within their own families in the main manuscript.

Contrast	$\Delta\rho$ [95% CI]	Won	p (p_{Holm})	Weighted $\Delta\rho$	Sign-flip p	N-B p
Hybrid-HGT vs HGT-QoS	+0.035 [−0.031, +0.108]	8/12	0.733	+0.083	0.339	0.536
Hybrid-GAT vs GAT-QoS	+0.048 [−0.019, +0.123]	7/12	0.301	+0.100	0.211	0.401
Hybrid-HGT vs Topo-QoS	+0.103 [+0.055, +0.151]	11/12	0.0034	+0.081	0.038	0.019
Hybrid-GAT vs Topo-QoS	+0.130 [+0.075, +0.188]	11/12	0.0015	+0.111	0.0049	0.014
Hybrid-HGT vs Topo (projection)	+0.065 [+0.014, +0.117]	9/12	0.042 (0.085)	+0.046	0.385	0.137
Hybrid-HGT vs Topo-Mult	+0.061 [+0.012, +0.116]	8/12	0.077 (0.085)	+0.046	0.368	0.167
Hybrid-GAT vs Topo (projection)	+0.092 [+0.034, +0.153]	10/12	0.012 (0.049)	+0.076	0.122	0.073
Hybrid-GAT vs Topo-Mult	+0.088 [+0.031, +0.151]	9/12	0.012 (0.049)	+0.075	0.121	0.092

was fixed in Amendment 14 after a preview of this very comparison (Amendment 14, “Status when written”), so it is not blind; the equivalence bound, the largest $|\cdot|$ of the 90% interval, is reported so that a reader can apply any margin.

Table S61: Top: GAT-P-QoS – InDeg on I^* per fold. Bottom: Nadeau–Bengio corrected resampled t (ratio 1/11; the Application-weighted ratio gives the same conclusions, see the artifact).

Statistic	Mean Δ	90% CI	TOST p (t / Wilcoxon / N-B)	Equivalence bound
Per-seed mean	−0.017	[−0.076, +0.043]	0.167 / 0.311 / 0.249	0.076
Seed ensemble	+0.007	[−0.058, +0.072]	0.132 / 0.151 / 0.216	0.072
Contrast	Mean Δ	t	N-B p	
HGT-QoS vs Topo-QoS (plan, v5)	+0.085	0.97	0.351	
HGT vs Topo-QoS (plan, v5)	−0.002	−0.04	0.967	
HGT-QoS vs Topo-QoS (CPU)	+0.068	0.80	0.442	
Hybrid-HGT vs Topo-QoS (CPU)	+0.103	2.75	0.019	
Hybrid-GAT vs Topo-QoS (CPU)	+0.130	2.92	0.014	
GAT-P-QoS vs InDeg	−0.017	−0.35	0.735	

Hierarchical intervals. Table S62 resamples folds and then seeds within folds ($B = 10,000$). The intervals widen only slightly against the fold-only bootstrap of the main manuscript’s Table 6, except for HGT-P-QoS, whose seeds disagree most.

Table S62: Learned rows of the main manuscript’s Table 6: fold-only and hierarchical fold→seed 95% intervals, and the mean within-fold seed SD.

Engine	ρ	Fold-only CI	Fold→seed CI	Seed SD
HGT-QoS	0.622	[0.548, 0.690]	[0.536, 0.697]	0.099
GAT-QoS	0.635	[0.566, 0.695]	[0.567, 0.698]	0.024
Hybrid-HGT	0.657	[0.570, 0.734]	[0.574, 0.733]	0.053
Hybrid-GAT	0.683	[0.602, 0.754]	[0.603, 0.754]	0.042
GAT-P-QoS	0.748	[0.704, 0.787]	[0.702, 0.789]	0.030
HGT-P-QoS	0.514	[0.395, 0.620]	[0.373, 0.640]	0.254
GAT-P+InDeg	0.758	[0.671, 0.830]	[0.674, 0.832]	0.011

Degree leak. Removing the two degree columns leaves other features that track in-degree. Table S63 gives the mean over folds of each Application feature’s Spearman correlation with InDeg; the strict arms of Amendment 14 also drop the three centralities above 0.7.

The $n = 30$ sensitivity check (Amendment 11, rule R). Table S64 sets the full-population I_{dyn} values of the main manuscript’s Table 7 beside the $n = 30$ lexicographic sample. Every ranker scores higher on the full population, the order of the rankers is unchanged, and I^* agrees with I_{dyn} at 0.711 against 0.627 on the sample.

The registered selection rule (arm N). Table S65 gives, per outer fold, the configuration the two-fold inner holdout selected and the outer ρ it reached, beside the published fixed configuration (3 layers, no rank normalization). The search harness early-stops on a held-out training scenario, whereas every published sweep early-stops on a 20% node-level split of the largest training scenario; the gate, which compares folds where the published hyperparameters were selected, fails for HGT-QoS (up to 0.102 apart on 3 folds), so the

Table S63: Mean Spearman ρ of Application node features with InDeg, twelve folds (features with $|\rho| \geq 0.15$).

Feature	ρ with InDeg
in_degree centrality	0.776
pagerank	0.775
closeness centrality	0.770
qos_weight_in	0.747
eigenvector centrality	0.738
betweenness centrality	0.452
mpci	0.448
code_quality_penalty	0.198
lcom_norm	0.188
complexity_norm	0.176
cdi	0.171

Table S64: Mean Spearman ρ against I_{dyn} : full population (five-seed mean) and the $n = 30$ lexicographic sample (seed 42), twelve LOSO folds.

Ranker	Full population	$n = 30$ sample	Partial I^* (full)
Analytic I^*	0.706	0.636	0.318
InDeg	0.664	0.610	0.272
Reach	0.583	0.505	0.117
Pubs-raw	0.645	0.571	0.255
Topo-QoS	0.471	0.393	0.134
Degree-raw	0.232	0.233	0.149
GAT-P-QoS	0.615	0.519	0.111
Hybrid-GAT	0.599	0.502	0.173
Hybrid-HGT	0.573	0.457	0.162
HGT-QoS	0.549	0.440	0.130
GAT-QoS	0.523	0.433	0.107
HGT-P-QoS	0.496	0.418	0.112

table compares two protocols, not two hyperparameter choices. Family F6 (Holm across three): nested vs. fixed HGT-QoS +0.055; nested vs. fixed GAT-P-QoS -0.054 (both $p_{\text{Holm}} = 0.259$); nested HGT-QoS vs. Topo-QoS +0.123 ($p_{\text{Holm}} = 0.192$, Nadeau-Bengio $p = 0.160$).

Cost reconciliation. Table S66 gives the per-fold rows behind the corpus row of the main manuscript’s Table 10. Every region was timed on the same topology in one session on an otherwise idle machine (CPU, one thread, median of three runs). The ratio of Table S29, measured in a separate session, compared the system-layer gate with the three-type sweep; the “Gate : sweep” column repeats that comparison in this session (1.7–16.3 $\times$, against 2.0–17.7 $\times$ in that session).

S42. Amendment 15: The Rate-Weighted Reference for I_{dyn} and Input Attribution

Amendment 15 is post hoc and exploratory: it was recorded after every other result existed, and its headline value was first computed ad hoc during the manuscript revision. It runs no simulator. `reproduce/idyn_rate_expansion.py` (make `rq-rate-expansion`) reads the five-seed, full-population I_{dyn} labels cached by Amendment 11 and writes `data/benchmarks/idyn_rate_expansion.json`. Two gates hold: the first-order expansion of I^* (main Eq. (6)) reproduces Amendment 11’s comparator on every fold, system and oracle ($\max |\Delta| = 0$), and the attribution arm with all Q columns reproduces the published GBM-P-QoS $\rightarrow$ dyn (0.799).

Closed forms. Table S67 scores four closed forms against I_{dyn} : the unweighted expansion (main Eq. (6)); the rate-weighted expansion (main Eq. (7)), the reference for I_{dyn} ; a payload-weighted variant ($r_t B_t$ in place of r_t); and the bare declared publication rate of v . A saturation-aware variant, adding lost flow at brokers whose declared capacity is exceeded, was not built: I_{dyn} runs at target utilization 0.65, and the plain rate form already matches or exceeds the learned approximation. Each closed form takes at most 1.2 ms per architecture.

Input attribution. The same table retrains Amendment 11’s gradient-boosted learner (same seeds, LOSO protocol and I_{dyn} labels) on the structural feature set S alone, on S plus the declared rate and rate $\times$ payload columns, and on S plus the seven QoS-derived columns: three $w(t)$ -weighted scores (Topo-QoS, Reach-QoS,

Table S65: Arm N: per-fold outer ρ on I^* under nested stage-1 selection, against the published fixed configuration. L: layers; RN-feat/RN-lab: rank-normalized features/labels.

Engine	Held-out fold	Fixed	Nested	Selected
HGT-QoS	atm	0.523	0.512	2L, RN-feat, RN-lab
HGT-QoS	av	0.704	0.727	3L, RN-feat
HGT-QoS	enterprise	0.426	0.634	3L, RN-feat, RN-lab
HGT-QoS	financial trading	0.695	0.752	2L, RN-feat
HGT-QoS	healthcare	0.730	0.805	2L, RN-feat
HGT-QoS	hub and spoke	0.548	0.546	2L, RN-feat, RN-lab
HGT-QoS	industrial scada	0.684	0.674	3L
HGT-QoS	iot smart city	0.688	0.821	2L, RN-feat, RN-lab
HGT-QoS	logistics fleet	0.771	0.755	3L, RN-feat
HGT-QoS	microservices	0.475	0.498	3L
HGT-QoS	realtime gaming	0.789	0.687	3L
HGT-QoS	telecom ran	0.427	0.708	3L, RN-feat
GAT-P-QoS	atm	0.603	0.471	2L, RN-feat
GAT-P-QoS	av	0.764	0.781	3L, RN-feat
GAT-P-QoS	enterprise	0.738	0.599	2L, RN-lab
GAT-P-QoS	financial trading	0.800	0.814	3L, RN-feat
GAT-P-QoS	healthcare	0.863	0.871	3L, RN-feat
GAT-P-QoS	hub and spoke	0.758	0.570	3L, RN-feat
GAT-P-QoS	industrial scada	0.834	0.718	2L, RN-lab
GAT-P-QoS	iot smart city	0.747	0.583	2L, RN-lab
GAT-P-QoS	logistics fleet	0.774	0.734	2L, RN-lab
GAT-P-QoS	microservices	0.649	0.677	2L, RN-lab
GAT-P-QoS	realtime gaming	0.795	0.826	2L, RN-feat
GAT-P-QoS	telecom ran	0.647	0.675	2L, RN-lab
Mean HGT-QoS / GAT-P-QoS		0.622 / 0.748	0.677 / 0.693	

Table S66: Per-fold like-for-like cost (seconds unless stated).

Fold	Count (ms)	One pass	Sweep	Features	Gate	Features : pass	Gate : sweep
ATM	0.4	0.01	0.09	0.07	0.16	4.5×	1.7×
AV System	2.7	0.10	0.40	1.87	2.64	19.5×	6.5×
Enterprise	15.6	0.72	4.98	52.25	81.48	72.5×	16.3×
Financial Trading	2.2	0.04	0.26	1.22	1.81	32.8×	6.9×
Healthcare	0.9	0.03	0.17	0.65	0.95	23.6×	5.7×
Enterprise Integration (ESB)	2.2	0.04	0.27	2.32	3.28	60.9×	12.2×
Industrial SCADA	1.4	0.25	1.31	1.34	2.58	5.4×	2.0×
IoT Smart City	3.5	0.22	1.24	3.19	5.81	14.3×	4.7×
Logistics Fleet	1.9	0.16	1.02	1.68	2.88	10.5×	2.8×
Microservices	2.1	0.13	0.91	1.49	2.42	11.2×	2.7×
Real-Time Gaming	1.3	0.06	0.42	1.60	2.57	28.9×	6.1×
Telecom RAN	2.5	0.19	1.13	2.30	3.83	12.2×	3.4×

QoS-weighted in-degree), in which declared size and rate enter log-compressed at a quarter of $w(t)$, and four pure policy shares (reliable, durable and deadline shares, maximum priority). Table S69 gives the paired contrasts.

Eq. (7) as the first wave of I_{dyn} . The derivation follows the engine’s delivery accounting (`saag/simulation/message_flow_simulator.py`). Each of the $|\text{pub}(t)|$ publishers of topic t emits at $r_t/|\text{pub}(t)|$, so that the topic’s aggregate rate is its declared r_t . A failed publisher keeps generating demand but emits nothing, so its messages stay in the expected-delivery count and leave only the delivered count. For one window, the delivery rate is delivered divided by expected deliveries, both taken over the subscribers other than v . Its expected-delivery rate is $D_v = \sum_t r_t |\text{sub}(t) \setminus \{v\}|$. Without queueing, deadlines or drops (S3), the pre-fault rate is 1, and the post-fault rate falls by the deliveries v ’s silence removes, so

$$I_{\text{dyn}}(v) = \frac{1}{D_v} \sum_{t \in \text{pub}(v)} \frac{r_t}{|\text{pub}(t)|} |\text{sub}(t) \setminus \{v\}|.$$

Because no failure propagates beyond one hop, this single wave is the whole effect. Dropping the per-component normalization $1/D_v$ (S4) and the exclusion of v from its own topics’ subscribers gives Eq. (7). D_v differs between the Applications of one architecture only by v ’s own subscriptions, so that simplification changes little of the within-architecture ranking. What the engine adds on top of this wave, bounded queues, service contention, deadlines and the RELIABLE/BEST_EFFORT drop policies, is what S3 omits; it is also why a measured pre-fault rate can lie below 1 and why I_{dyn} can be negative when removing a publisher relieves contention.

Payload is not read by the oracle. The published I_{dyn} engine gives every message the same size, so declared payload never affects its labels (main Section 4.3). The rate $\times$ payload column can therefore carry signal only through its rate factor, and the gain of the “S + rate, payload” arm is rate signal. Consistently, the payload-weighted closed form ($r_t B_t$) scores below Eq. (7) on all twelve folds (mean 0.748 against 0.830). A payload-aware variant of the oracle is registered as Amendment 18 but was not run.

Table S67: Amendment 15 on I_{dyn} (five-seed mean, full population), twelve LOSO folds. Closed forms are training-free; the learned columns are means over five seeds. GBM $\rightarrow$ dyn is Amendment 11’s GBM-P-QoS $\rightarrow$ dyn (S + all Q).

Fold	Eq. (6)	Closed forms			Learned approximation (GBM, trained on I_{dyn})			
		Eq. (7)	$r_t B_t$	Rate	GBM $\rightarrow$ dyn	S	S + rate, payload	S + QoS-derived
ATM	0.533	0.838	0.628	0.770	0.798	0.644	0.842	0.586
AV	0.634	0.741	0.732	0.748	0.742	0.649	0.736	0.628
Enterprise	0.859	0.856	0.801	0.827	0.824	0.831	0.824	0.834
Financial Trading	0.833	0.847	0.758	0.854	0.879	0.836	0.876	0.861
Healthcare	0.898	0.898	0.841	0.834	0.893	0.913	0.887	0.903
Hub-and-Spoke	0.481	0.598	0.516	0.517	0.590	0.504	0.560	0.524
Industrial SCADA	0.602	0.759	0.708	0.681	0.732	0.620	0.727	0.618
IoT Smart City	0.923	0.923	0.843	0.855	0.865	0.891	0.882	0.885
Logistics Fleet	0.758	0.886	0.814	0.849	0.846	0.722	0.847	0.741
Microservices	0.660	0.901	0.735	0.823	0.863	0.633	0.862	0.603
Real-Time Gaming	0.690	0.866	0.821	0.831	0.763	0.612	0.769	0.604
Telecom RAN	0.601	0.850	0.782	0.839	0.797	0.597	0.801	0.608
Mean	0.706	0.830	0.748	0.786	0.799	0.704	0.801	0.700

Table S68: Amendment 15, zero-shot on the five system models, I_{dyn} . GBM $\rightarrow$ dyn is trained on the twelve synthetic folds.

System	Eq. (6)	Eq. (7)	$r_t B_t$	Rate	GBM $\rightarrow$ dyn
Autoware	0.424	0.839	0.576	0.537	0.668
EdgeX	0.844	0.926	0.924	0.750	0.834
Home Assistant	0.723	0.972	0.909	0.890	0.790
Online Boutique	0.775	0.815	0.751	0.673	0.806
Train-Ticket	0.841	0.911	0.906	0.819	0.899
Mean	0.722	0.893	0.813	0.734	0.800

Table S69: Amendment 15 contrasts on I_{dyn} , twelve LOSO folds. Two-sided Wilcoxon signed-rank; Holm within each block. Nominal: folds share training scenarios, and the amendment is post hoc.

Contrast	$\Delta\rho$ [95% CI]	Won	p	p_{Holm}
Eq. (7) vs GBM $\rightarrow$ dyn	+0.031 [+0.013, +0.049]	10/12	0.009	0.009
Eq. (7) vs Eq. (6)	+0.124 [+0.068, +0.184]	9/12	0.004	0.008
Eq. (7) vs InDeg	+0.166 [+0.091, +0.245]	11/12	0.002	0.007
S + rate, payload vs S	+0.097 [+0.049, +0.143]	9/12	0.007	0.021
S + QoS-derived vs S	-0.005 [-0.019, +0.007]	5/12	0.677	1.000
S + all Q vs S + rate, payload	-0.002 [-0.012, +0.007]	6/12	0.970	1.000

The rate-weighted expansion keeps a partial correlation of 0.578 with I_{dyn} after the rank of I^* is removed, against 0.318 for the unweighted expansion: the rate term is exactly the part of I_{dyn} that I^* does not contain. On the other oracles it scores 0.756 (I^*) and 0.551 (I_{comp}).

S43. Control Arms (Amendments 14, 16, 17 and 19)

Main Table 8 is a digest of the tables below, which give every registered control arm with its I_{dyn} , I_{comp} and zero-shot values. Each table was produced in one CPU invocation in which every comparator was re-run and reproduced its published per-seed ρ . Section 6.2 of the main manuscript interprets them.

Notes moved from the main text.

- **Degree features (F1).** Also removing the three centralities most correlated with in-degree (PageRank, closeness and eigenvector centrality) lowers GAT-P-QoS by 0.161; the GINE network of the same size keeps 0.711 without those columns. The aggregator contrasts of F2 are not significant after Holm correction (+0.108 and +0.124, $p_{\text{Holm}} = 0.157$).
- **Hybrid priors (F9, F10).** Retrained with the articulation defect corrected, Hybrid-GAT-AP reaches 0.669 (+0.136, Holm $p = 0.0059$) and Hybrid-HGT-AP 0.640 (+0.107, Holm $p = 0.0073$), each on 11 of 12 folds, and again neither differs from its base learner (+0.034 and +0.018, Holm $p = 0.94$). Given InDeg as their prior, GAT-QoS+InDeg (0.763) and HGT-QoS+InDeg (0.759) gain +0.128 and +0.137 over their base learners (Holm $p = 0.0049$) and land within ± 0.012 of InDeg itself (equivalent at ± 0.05 , TOST $p < 0.001$).
- **Learned queue-flow approximation.** The GAT trained on I_{dyn} labels (0.598) falls below the unweighted first-order expansion (-0.108, Holm $p = 0.014$) and is no better than the same GNN trained on I^* labels restricted to Applications, a label-support control run in the same sweep (-0.008).
- **Node order (F13).** Creation order correlates only weakly with the labels ($\rho = -0.20$ to +0.29 per fold), and each permutation also changes which nodes the seeded validation split draws, so the spread across permutations mixes tie order and validation draw. Differences such as the -0.016 between GIN-P-QoS and GAT-P-QoS lie within it.
- **Nested selection.** On the three folds where the nested harness chose the published hyperparameters it does not reproduce the published scores (differences up to 0.10), because it trains on one scenario fewer and stops on a held-out scenario. Under that protocol GAT-P-QoS (0.693) falls 0.071 [0.002, 0.143] below InDeg on 9 of 12 folds ($p = 0.11$).
- **Mixed effects (Amendment 19).** A linear mixed model of per-seed ρ on the arm, with fold as group, a random arm slope and seeds as residual, gives the same estimates as the paired fold-level tests: +0.072 ($p < 0.001$) for GAT-P-QoS against GAT-QoS-R, +0.231 and +0.239 ($p < 0.001$) for the dependency graph against the reverse-edge controls without oracle-aligned features under attention and under sum aggregation, +0.069 ($p = 0.002$) under the tie-aware loss, and -0.017 ($p = 0.62$) for GAT-P-QoS against InDeg. All models converged.

- **Tie-aware loss and node order (Amendment 19).** With tied labels as groups, the three node-order permutations of GAT-P-QoS-tie give 0.711, 0.741 and 0.738, a mean per-fold spread of 0.047 against 0.044 under ListMLE, so the order dependence does not come mainly from tie order (rule S2).
- **Capacity (Amendment 19).** At width 64 (31,048 parameters), GAT-S-P-QoS reaches 0.638, 0.110 below GAT-P-QoS on every fold: the published width is not over-parameterized for this task.

Table S70: Degree and aggregator arms: twelve LOSO folds, five seeds, one CPU invocation with every comparator re-run to an exact match, Application population. $\Delta\rho$ paired by fold with a bootstrap 95% CI; Holm within each registered family (F1: degree features; F2: aggregator; F4: the 2×2 with w_{in} held). -deg: in_degree and w_{in} zeroed; -deg*: also PageRank, closeness and eigenvector centrality. Zero-shot: mean ρ on the five system models. Registered secondary.

Arm	Comparator	LOSO ρ (I^*)	$\Delta\rho$ [95% CI]	Won	p_{Holm}	I_{dyn} / I_{comp}	Zero-shot
<i>F1: removing the degree features</i>							
GAT-P-QoS-deg	GAT-P-QoS (0.748)	0.612	-0.136 [-0.205, -0.072]	1/12	0.0049	0.485 / 0.156	0.821
GAT-P-QoS-deg*	GAT-P-QoS	0.587	-0.161 [-0.252, -0.080]	1/12	0.0049	0.462 / 0.138	0.654
GAT-QoS-deg	GAT-QoS (0.635)	0.365	-0.270 [-0.344, -0.199]	0/12	0.0015	0.295 / -0.047	0.810
<i>F2: sum aggregation (Graph Isomorphism Network with Edge features — GINE, 434,123 parameters) instead of softmax attention</i>							
GIN-P-QoS	GAT-P-QoS	0.732	-0.016 [-0.055, +0.017]	7/12	0.910	0.606 / 0.407	0.806
GIN-P-QoS-deg	GAT-P-QoS-deg	0.721	+0.108 [+0.027, +0.197]	8/12	0.157	0.602 / 0.428	0.810
GIN-P-QoS-deg*	GAT-P-QoS-deg*	0.711	+0.124 [+0.031, +0.230]	9/12	0.157	0.589 / 0.438	0.776
<i>F4: matched 2×2 with w_{in} held (cells GAT+w_{in} 0.628, HGT+w_{in} 0.569, GAT-QoS 0.635, HGT-QoS 0.622)</i>							
Typing (main effect)	averaged over Q	—	-0.036 [-0.079, +0.004]	2/12	0.330	—	—
“QoS” inputs (main effect)	averaged over T	—	+0.030 [-0.023, +0.085]	9/12	0.330	—	—
Interaction	difference of differences	—	+0.046 [-0.009, +0.101]	9/12	0.330	—	—

Table S71: Direction and prior controls: twelve LOSO folds, five seeds, one CPU invocation, Application population; $\Delta\rho$ paired by fold with a bootstrap 95% CI, Holm within each registered family (F8: direction; F9: corrected prior; F10: InDeg prior). GAT-QoS-R: GAT-QoS with every raw-graph edge also passed in reverse (shared weights, 429,992 parameters). -AP: prior with the articulation term restored; Topo-QoS-AP is the corrected baseline. Re-run comparators reproduce their published per-seed ρ exactly, except Hybrid-HGT (within 1.2×10^{-4} ; not a comparator here). Zero-shot: mean ρ on the five system models. Registered secondary.

Arm	Comparator	LOSO ρ (I^*)	$\Delta\rho$ [95% CI]	Won	p_{Holm}	I_{dyn} / I_{comp}	Zero-shot
<i>F8: edge direction versus dependency semantics</i>							
GAT-QoS-R	GAT-QoS (0.635)	0.676	+0.041 [+0.000, +0.087]	10/12	0.064	0.547 / 0.299	0.744
GAT-P-QoS	GAT-QoS-R	0.748	+0.072 [+0.034, +0.113]	10/12	0.014	0.597 / 0.280	—
<i>F9: hybrids with the corrected prior</i>							
Hybrid-GAT-AP	Topo-QoS-AP (0.533)	0.669	+0.136 [+0.076, +0.198]	11/12	0.0059	0.577 / 0.561	0.668
Hybrid-HGT-AP	Topo-QoS-AP	0.640	+0.107 [+0.060, +0.153]	11/12	0.0073	0.552 / 0.553	0.702
Hybrid-GAT-AP	GAT-QoS	0.669	+0.034 [-0.047, +0.118]	7/12	0.940	—	—
Hybrid-HGT-AP	HGT-QoS (0.622)	0.640	+0.018 [-0.067, +0.100]	8/12	0.940	—	—
<i>F10: hybrids with the InDeg prior (within ± 0.012 of InDeg, 0.764)</i>							
GAT-QoS+InDeg	GAT-QoS	0.763	+0.128 [+0.063, +0.204]	11/12	0.0049	0.657 / 0.563	—
HGT-QoS+InDeg	HGT-QoS	0.759	+0.137 [+0.074, +0.211]	10/12	0.0049	0.657 / 0.589	—

S44. Training-Free Baselines

This section consolidates the non-registered training-free baselines and controls moved out of the main manuscript. The main manuscript retains Topo-QoS as its sole training-free baseline, because it forms one side of the registered primary contrast and acts as the prior inside both hybrid engines.

Unweighted Topo Baseline. The unweighted topological baseline Topo is evaluated strictly on the raw application layer without Rule 5 library derivation:

$$\text{Topo}(v) = 0.6 \cdot \text{BT}(v) + 0.4 \cdot \text{AP}(v), \quad (7)$$

Table S72: Oracle-aligned-feature, learning-on-Eq. (7) and node-order controls: twelve LOSO folds, five seeds, one CPU invocation with every comparator re-run to an exact match, Application population. $\Delta\rho$ is paired by fold against the comparator named, with a bootstrap 95% CI; Holm within each registered family (F11: oracle-aligned features; F12: learners started from Eq. (7)). **-min**: in-degree, w_{in} , reverse PageRank, the articulation score, multi-path coupling, fan-out criticality and CDI zeroed; **-const**: every node feature zeroed. The F12 rows report ρ on I_{dyn} and are trained on its labels; all other rows report ρ on I^* . F13 is a registered gate with a nominal p . I_{dyn} / I_{comp} : per-seed means from saved predictions. Zero-shot: mean ρ on the five system models (— where not run). Registered secondary.

Arm	Comparator	ρ	$\Delta\rho$ [95% CI]	Won	p_{Holm}	I_{dyn} / I_{comp}	Zero-shot
<i>F11: every oracle-aligned feature removed (registered)</i>							
GAT-P-QoS-min	GAT-QoS-R-min (0.378)	0.610	+0.231 [+0.128, +0.340]	10/12	0.0068	0.479 / 0.149	0.839
GAT-P-QoS-min	GAT-P-QoS (0.748)	0.610	-0.138 [-0.208, -0.072]	1/12	0.0044	—	—
GIN-P-QoS-min	GAT-P-QoS-min	0.724	+0.115 [+0.038, +0.199]	9/12	0.027	0.606 / 0.423	0.802
<i>Descriptive (no test)</i>							
GAT-QoS-R-min	GAT-QoS-R (0.676)	0.378	-0.298 [-0.387, -0.222]	0/12	—	0.291 / 0.319	0.708
GAT-QoS-min	GAT-QoS (0.635)	0.369	-0.266 [-0.334, -0.201]	0/12	—	0.294 / -0.049	0.794
GIN-P-QoS-const	InDeg (0.764)	0.719	-0.045 [-0.069, -0.024]	1/12	—	0.606 / 0.447	0.858
<i>F12: learners started from Eq. (7), scored on I_{dyn} (registered)</i>							
GBM-P-QoS→dyn+Eq7	Eq. (7) (0.830)	0.830	+0.000 [-0.013, +0.015]	5/12	0.970	—	—
GBM→dyn-resid	Eq. (7)	0.824	-0.006 [-0.019, +0.008]	5/12	0.679	—	—
GAT-P-QoS→dyn+Eq7	Eq. (7)	0.812	-0.018 [-0.025, -0.012]	1/12	0.0029	—	—
<i>F13: node order permuted before training (registered gate; nominal p)</i>							
GAT-P-QoS-perm	GAT-P-QoS	0.712	-0.035 [-0.055, -0.014]	3/12	(0.012)	0.578 / 0.227	—

Table S73: Amendment 19 control arms: twelve LOSO folds, five seeds, one CPU invocation with every comparator re-run to an exact match, Application population. $\Delta\rho$ is paired by fold against the comparator named, with a bootstrap 95% CI; Holm within each registered family (F14: aggregator; F15: rate-fed approximations against Eq. (7); F16: tie-aware loss). **GIN**: sum aggregation (GINE layers); **-R**: every raw-graph edge also passed in reverse; **-min**: oracle-aligned features zeroed; **-const**: every node feature zeroed; **+rate**: each node's summed declared publication rate as an input column; **+rate-e**: also each Rule-1 edge's share of Eq. (7) as an edge column; **-tie**: tie-aware listwise loss; **GAT-S-P-QoS**: width 64. F15 rows report ρ on I_{dyn} ; all others on I^* . I_{dyn} / I_{comp} : per-seed means from saved predictions. Zero-shot: mean ρ on the five system models (— where not run). Mean per-fold spread across three node-order permutations of **GAT-P-QoS-tie**: 0.047 (ListMLE: 0.044). Decision rules: F14 F14a, F15 F15b, F16 F16b-holds, S S2. Registered secondary.

Arm	Comparator	ρ	$\Delta\rho$ [95% CI]	Won	p_{Holm}	I_{dyn} / I_{comp}	Zero-shot
<i>F14: sum aggregation with reverse edges on the raw multigraph (registered)</i>							
GIN-P-QoS-min	GIN-QoS-R-min (0.485)	0.724	+0.239 [+0.151, +0.335]	12/12	0.0015	0.606 / 0.423	—
GIN-P-QoS	GIN-QoS-R (0.668)	0.732	+0.064 [+0.031, +0.099]	10/12	0.0049	0.606 / 0.407	—
GIN-P-QoS-const	GIN-QoS-R-const (0.430)	0.719	+0.289 [+0.191, +0.387]	11/12	0.0020	0.606 / 0.447	—
<i>Descriptive (no test)</i>							
GIN-QoS-R	GAT-QoS-R (0.676)	0.668	-0.008 [-0.061, +0.042]	8/12	—	0.575 / 0.497	0.766
GIN-QoS-R-min	GAT-QoS-R-min (0.378)	0.485	+0.107 [+0.019, +0.191]	9/12	—	0.448 / 0.480	0.725
GIN-QoS-R-const	InDeg (0.764)	0.430	-0.334 [-0.440, -0.238]	0/12	—	0.424 / 0.617	0.399
<i>F15: rate-fed learned approximations, scored on I_{dyn} (registered)</i>							
GAT-P-QoS→dyn+rate	Eq. (7) (0.830)	0.622	-0.208 [-0.263, -0.151]	0/12	0.0015	—	—
GAT-P-QoS→dyn+rate-e	Eq. (7) (0.830)	0.625	-0.205 [-0.260, -0.148]	0/12	0.0015	—	—
GIN-P-QoS→dyn+rate-e	Eq. (7) (0.830)	0.665	-0.165 [-0.231, -0.102]	1/12	0.0015	—	—
GAT-P-QoS→dyn+rate	GAT-P-QoS→dyn (0.598)	0.622	+0.024 [+0.007, +0.044]	9/12	—	—	—
GAT-P-QoS→dyn+rate-e	GAT-P-QoS→dyn (0.598)	0.625	+0.027 [+0.012, +0.044]	11/12	—	—	—
GIN-P-QoS→dyn+rate-e	GAT-P-QoS→dyn (0.598)	0.665	+0.067 [+0.019, +0.107]	10/12	—	—	—
<i>F16: tie-aware listwise loss (registered)</i>							
GAT-P-QoS-tie	GAT-P-QoS (0.748)	0.747	-0.001 [-0.008, +0.005]	6/12	0.970	0.595 / 0.272	—
GAT-P-QoS-tie	GAT-QoS-R-tie (0.677)	0.747	+0.069 [+0.031, +0.115]	9/12	0.019	0.595 / 0.272	—
<i>Small-capacity arm (descriptive)</i>							
GAT-S-P-QoS	GAT-P-QoS (0.748)	0.638	-0.110 [-0.147, -0.073]	0/12	—	0.515 / 0.233	—
GAT-S-P-QoS	InDeg (0.764)	0.638	-0.126 [-0.189, -0.066]	2/12	—	0.515 / 0.233	—

Table S74: Learning curve (Amendment 19): LOSO Spearman ρ against I^* with each fold’s learner trained on K of its eleven training scenarios (mean over three nested subset draws and five seeds; $K = 11$ is the full corpus). $\Delta(11-4)$ is paired by fold, Holm across the three learners; $\Delta(11-8)$ is descriptive. Afferent coupling (**InDeg**) reaches 0.764 without training. Overall rule: LC-c. Registered secondary.

Learner	$K = 1$	$K = 2$	$K = 4$	$K = 8$	$K = 11$	$\Delta(11-4)$ [95% CI]	p_{Holm}	$\Delta(11-8)$ [95% CI]	Rule
GAT-P-QoS	0.605	0.648	0.670	0.732	0.748	+0.078 [+0.031, +0.142]	0.015	+0.015 [+0.003, +0.028]	LC-c
GIN-P-QoS	0.645	0.678	0.705	0.730	0.732	+0.027 [+0.010, +0.044]	0.024	+0.001 [−0.012, +0.021]	LC-c
GAT-QoS	0.579	0.611	0.622	0.632	0.635	+0.013 [−0.002, +0.032]	0.176	+0.003 [−0.004, +0.010]	LC-a

where $\text{BT}(v)$ is normalized betweenness centrality, and $\text{AP}(v)$ flags articulation points. Due to an implementation defect in the registered benchmark code (disclosed in Section 5.2 of the main manuscript), the articulation point term reads zero for every node, so the reported score ranks nodes identically to betweenness centrality on the application layer. This distinguishes **Topo** from **Topo-QoS**, which evaluates QoS-weighted betweenness on the derived dependency graph.

Performance on Synthetic Architectures. Table S75 reports the performance of **Degree-raw**, **RevPR-raw**, **Topo**, and the raw-multigraph reference rankings (**Pubs-raw**, **Reach-R1**) across the twelve synthetic LOSO folds.

Table S75: Non-registered training-free baselines and exploratory raw-graph references evaluated under Leave-One-Scenario-Out (LOSO) cross-validation across twelve synthetic architectures (Application population). $\Delta\rho$ is paired by fold against **Topo-QoS** with a bootstrap 95% CI; two-sided Wilcoxon signed-rank test against **Topo-QoS**.

Predictor	LOSO ρ [95% CI]	Active $\rho_{>0}$	$\Delta\rho$ vs Topo-QoS [95% CI]	Won	p	Overlap@ K
<i>Training-free, raw multigraph (no derivation)</i>						
Degree-raw	0.199 [0.068, 0.328]	0.218	−0.354 [−0.475, −0.236]	1/12	0.0015	0.307
RevPR-raw	0.089 [−0.019, 0.192]	0.075	−0.465 [−0.558, −0.364]	0/12	0.0005	0.199
<i>Training-free, application layer</i>						
Topo	0.349 [0.254, 0.452]	0.174	−0.204 [−0.286, −0.122]	0/12	0.0005	0.366
<i>Raw-multigraph reference rankings (Amendment 12)</i>						
Pubs-raw	0.731 [0.637, 0.810]	0.410	—	—	—	0.487
Reach-R1	0.674 [0.606, 0.738]	0.088	—	—	—	0.300

Where the closed-form gain comes from. **Topo** reads betweenness from the application-layer graph, whereas **Topo-QoS** computes QoS-weighted betweenness on the Application–Library dependency graph. Three controls registered in Amendment 7 separate substrate from weighting: unweighted betweenness on the same graph scores 0.591; constant topic weights score 0.595; permuting QoS profiles scores 0.559 (−0.006, $p = 0.73$). On I^* , the gain comes from the Application–Library graph structure, not from QoS contract content.

Against the two stronger training-free controls of Amendment 7 (Amendment 14, F7), Hybrid-GAT keeps a margin (unweighted betweenness on the same graph +0.092, constant topic weights +0.088, both Holm $p = 0.049$), Hybrid-HGT does not (+0.065 and +0.061, Holm $p = 0.085$).

Degree-raw and Supplemental Predictors Across Simulation Oracles. Table S76 reports supplemental predictors and baselines across the three simulation oracles. On the multi-criteria oracle I_{comp} , raw total degree reaches $\rho = 0.719$, exceeding every learned engine without a prior.

Topo on Open-Source System Models. Table S77 reports the zero-shot performance of unweighted **Topo** on the five hand-authored open-source system models.

Table S76: Supplemental predictors and baselines against three simulation oracles, twelve LOSO folds, Application population. Partial ρ : Spearman correlation with I_{dyn} after I^* or Analytic I^* ($\hat{I}_1^*$) is regressed out.

Predictor	I^*	I_{dyn} [95% CI]	Partial vs I^*	Partial vs $\hat{I}_1^*$	I_{comp}
Degree-raw	0.199	0.232 [0.134, 0.329]	0.149 [0.089, 0.227]	−0.091	0.719
Pubs-raw	0.731	0.645 [0.543, 0.737]	0.255 [0.179, 0.332]	0.056	0.616
GBM-P-QoS	0.818	0.718	—	—	0.483

Table S77: Unweighted Topo evaluated zero-shot across the five hand-authored open-source system models (Application population).

Predictor	Substrate	LOSO ρ [95% CI]	Active $\rho_{>0}$	PR-AUC
Topo	Application layer	0.511 [0.346, 0.703]	−0.104	0.474